

Data Analyst Interview Bible (UK 2026)

UK Interview Edition

Bohdan Shevchuk, ChatGPT (OpenAI)

Contents

1	evenec	29
2	Data Analyst Interview Bible (UK 2026)	31
2.1	Part 1 – HR Interview Mastery	31
2.2	Part 2 – SQL Bible	31
3		33
4		35
I	Part I – HR Interview Mastery	
5	HR Interview Mastery	39
6	Questions 1–15	41
7	How to Use This Chapter	43
8	Question 1 – Tell me about yourself	45
8.1	Recruiter Psychology	45
8.2	Model Answer	45
8.2.1	Follow-up Questions	45
8.2.2	Common Mistake	45
9	Question 2 – Why do you want to become a Data Analyst?	47
9.1	Recruiter Psychology	47
9.2	Model Answer	47
9.2.1	Common Mistake	47
10	Question 3 – Why are you leaving your current job?	49
10.1	Recruiter Psychology	49
10.2	Model Answer	49
10.2.1	Follow-up Questions	49
10.2.2	Common Mistake	49
11	Practice Break	51
12	Question 4 – Why do you want to work for our company?	53
12.1	Recruiter Psychology	53
12.2	Model Answer	53
12.2.1	Company Research Checklist	53
12.2.2	Common Mistake	53

13	Question 5 – What are your strengths?	55
13.1	Recruiter Psychology	55
13.2	Model Answer	55
13.2.1	Memory Hook	55
13.2.2	Common Mistake	55
14	Question 6 – What is your biggest weakness?	57
14.1	Recruiter Psychology	57
14.2	Model Answer	57
14.2.1	Good Weaknesses	57
14.2.2	Common Mistake	57
15	Practice Break	59
16	Question 7 – Tell me about a challenge you faced.	61
16.1	Recruiter Psychology	61
16.2	Model Answer	61
16.2.1	Situation	61
16.2.2	Task	61
16.2.3	Action	61
16.2.4	Result	61
16.2.5	Bogdan Note	61
17	Question 8 – Tell me about a mistake you made.	63
17.1	Recruiter Psychology	63
17.2	Model Answer	63
17.2.1	Common Mistake	63
18	Question 9 – How do you prioritise your work?	65
18.1	Recruiter Psychology	65
18.2	Model Answer	65
18.2.1	Common Mistake	65
19	Practice Break	67
20	Question 10 – Tell me about working with difficult people. . . .	69
20.1	Recruiter Psychology	69
20.2	Model Answer	69
20.2.1	Bogdan Note	69
21	Question 11 – Describe teamwork.	71
21.1	Recruiter Psychology	71
21.2	Model Answer	71
21.2.1	Memory Hook	71
22	Question 12 – How do you handle pressure?	73
22.1	Recruiter Psychology	73
22.2	Model Answer	73
22.2.1	Common Mistake	73

23	Question 13 — Where do you see yourself in five years?	75
23.1	Recruiter Psychology	75
23.2	Model Answer	75
23.2.1	Common Mistake	75
24	Question 14 — Why should we hire you?	77
24.1	Recruiter Psychology	77
24.2	Model Answer	77
24.2.1	Memory Hook	77
25	Question 15 — Do you have any questions for us?	79
25.1	Recruiter Psychology	79
25.1.1	Model Answer	79
25.1.2	Five Excellent Questions	79
25.1.3	Common Mistake	79
26	Chapter Summary	81
26.1	Top Five Memory Hooks	81
27	5-Minute Interview Drill	83
28	HR Interview Mastery	85
29	Questions 16–30	87
30	Question 16 — Tell me about a time you learned something new quickly.	89
30.1	Recruiter Psychology	89
30.2	Model Answer	89
30.2.1	Common Mistake	89
31	Question 17 — Describe a time you improved a process.	91
31.1	Recruiter Psychology	91
31.2	Model Answer	91
31.2.1	Memory Hook	91
32	Question 18 — Tell me about a time you worked with data.	93
32.1	Recruiter Psychology	93
32.2	Model Answer	93
33	Practice Break	95
34	Question 19 — How do you deal with repetitive tasks?	97
34.1	Recruiter Psychology	97
34.2	Model Answer	97
34.2.1	Memory Hook	97
34.2.2	Common Mistake	97

35	Question 20 — Tell me about receiving feedback.	99
35.1	Recruiter Psychology	99
35.2	Model Answer	99
35.2.1	Memory Hook	99
36	Question 21 — Tell me about giving feedback.	101
36.1	Recruiter Psychology	101
36.2	Model Answer	101
36.2.1	Common Mistake	101
37	Question 22 — How do you organise your work?	103
37.1	Recruiter Psychology	103
37.2	Model Answer	103
38	Practice Break	105
39	Question 23 — Tell me about a time you solved a problem. . . .	107
39.1	Recruiter Psychology	107
39.2	Model Answer	107
39.2.1	Memory Hook	107
40	Question 24 — How do you stay motivated?	109
40.1	Recruiter Psychology	109
40.2	Model Answer	109
40.2.1	Common Mistake	109
41	Question 25 — What does success mean to you?	111
41.1	Recruiter Psychology	111
41.2	Model Answer	111
41.2.1	Memory Hook	111
42	Practice Break	113
43	Question 26 — Tell me about working under a deadline.	115
43.1	Recruiter Psychology	115
43.2	Model Answer	115
43.2.1	Memory Hook	115
44	Question 27 — Describe a situation where you took initiative. .	117
44.1	Recruiter Psychology	117
44.2	Model Answer	117
45	Question 28 — How do you deal with uncertainty?	119
45.1	Recruiter Psychology	119
45.2	Model Answer	119
45.2.1	Memory Hook	119

46	Question 29 — Describe a time you had multiple priorities. . . .	121
46.1	Recruiter Psychology	121
46.2	Model Answer	121
46.2.1	Common Mistake	121
47	Question 30 — Why should we believe you'll succeed as a Junior Data Analyst without previous analyst experience? . . .	123
47.1	Recruiter Psychology	123
47.2	Model Answer	123
47.2.1	Common Mistake	123
48	Chapter Summary	125
48.1	Five Key Memory Hooks	125
49	5-Minute Interview Drill	127
50	HR Interview Mastery	129
51	Questions 31–40	131
52	Question 31 — Tell me about a time you had to learn from failure.	133
52.1	Recruiter Psychology	133
52.2	Model Answer	133
52.2.1	Common Mistake	133
53	Question 32 — Describe a time you had to explain something complicated.	135
53.1	Recruiter Psychology	135
53.2	Model Answer	135
53.2.1	Memory Hook	135
53.2.2	Follow-up Questions	135
54	Question 33 — How do you handle changing priorities?	137
54.1	Recruiter Psychology	137
54.2	Model Answer	137
54.2.1	Common Mistake	137
55	Practice Break	139
56	Question 34 — Tell me about a time you had to make a difficult decision.	141
56.1	Recruiter Psychology	141
56.2	Model Answer	141
56.2.1	Memory Hook	141

57	Question 35 — What would you do if you didn't know the answer?	143
57.1	Recruiter Psychology	143
57.2	Model Answer	143
57.2.1	Common Mistake	143
58	Question 36 — Tell me about a time you worked independently.	145
58.1	Recruiter Psychology	145
58.2	Model Answer	145
58.2.1	Memory Hook	145
59	Practice Break	147
60	Question 37 — Describe a situation where you had to be very accurate.	149
60.1	Recruiter Psychology	149
60.2	Model Answer	149
60.2.1	Common Mistake	149
61	Question 38 — How do you react when someone disagrees with you?	151
61.1	Recruiter Psychology	151
61.2	Model Answer	151
61.2.1	Memory Hook	151
62	Question 39 — Tell me about a goal you achieved.	153
62.1	Recruiter Psychology	153
62.2	Model Answer	153
62.2.1	Common Mistake	153
63	Question 40 — Why should we choose you over another candidate?	155
63.1	Recruiter Psychology	155
63.2	Model Answer	155
63.2.1	Common Mistake	155
64	Chapter Summary	157
64.1	Five Key Memory Hooks	157
65	STAR Practice	159
66	End-of-Chapter Reflection	161
67	HR Interview Mastery	163
68	Questions 41–50	165

69	Question 41 — Tell me about a time you disagreed with a decision.	167
69.1	Recruiter Psychology	167
69.2	Model Answer	167
69.2.1	Common Mistake	167
70	Question 42 — Tell me about adapting to change.	169
70.1	Recruiter Psychology	169
70.2	Model Answer	169
70.2.1	Memory Hook	169
71	Question 43 — How do you handle constructive criticism? . . .	171
71.1	Recruiter Psychology	171
71.2	Model Answer	171
71.2.1	Memory Hook	171
72	Practice Break	173
73	Question 44 — Describe a situation where you helped someone.	175
73.1	Recruiter Psychology	175
73.2	Model Answer	175
73.2.1	Common Mistake	175
74	Question 45 — How do you stay organised when learning new skills?	177
74.1	Recruiter Psychology	177
74.2	Model Answer	177
75	Question 46 — Tell me about making a decision with limited information.	179
75.1	Recruiter Psychology	179
75.2	Model Answer	179
75.2.1	Memory Hook	179
76	Practice Break	181
77	Question 47 — What motivates you to keep learning?	183
77.1	Recruiter Psychology	183
77.2	Model Answer	183
77.2.1	Memory Hook	183
78	Question 48 — Describe a time you worked carefully with details.	185
78.1	Recruiter Psychology	185
78.2	Model Answer	185
78.2.1	Memory Hook	185

79	Question 49 — What would your previous colleagues say about you?	187
79.1	Recruiter Psychology	187
79.2	Model Answer	187
79.2.1	Common Mistake	187
80	Question 50 — Is there anything else you'd like us to know? . .	189
80.1	Recruiter Psychology	189
80.2	Model Answer	189
80.2.1	Final Impression Tip	189
81	Chapter Summary	191
81.1	Five Final Memory Hooks	191
82	Salary Negotiation (UK £35–45k)	193
82.1	When should you discuss salary?	193
82.2	If HR asks first	193
82.2.1	Best Answer (B2)	193
82.3	If they offer £32,000	193
82.3.1	Good Response	193
82.4	If they ask for your minimum salary	193
82.5	Salary Negotiation Checklist	194
83	Recruiter Red Flags	195
83.1	Green Flags	195
83.2	Yellow Flags	195
83.3	Red Flags	195
84	Mini Mock Interview	197
84.1	Interview Starts	197
85	Self-Evaluation	199
86	Final Interview Checklist	201
86.1	The Day Before	201
86.2	One Hour Before	201
86.3	During the Interview	201
86.4	After the Interview	201
87	Part 1 Complete	203

II

Part II — SQL Bible

88	SQL Bible	207
-----------	----------------------------	------------

89	Sprint 1 — Foundations	209
89.1	SELECT • WHERE • ORDER BY • LIMIT • DISTINCT	209
90	Introducing the Evenec Retail Dataset	211
90.1	Database Structure	211
91	Main Tables	213
91.1	customers	213
91.2	orders	213
91.3	products	213
92	Memory Map	215
93	Question 1 — Show Every Customer	217
93.1	Recruiter Psychology	217
93.2	Business Task	217
93.3	SQL Solution	217
93.4	Explain It (B2)	217
93.5	Memory Hook	217
93.6	Common Mistake	217
93.7	Practice Challenge	217
94	Question 2 — Select Specific Columns	219
94.1	Recruiter Psychology	219
94.2	Business Task	219
94.3	SQL Solution	219
94.4	Explain It (B2)	219
94.5	Memory Hook	219
94.6	Common Mistake	219
95	Question 3 — Filter London Customers	221
95.1	Recruiter Psychology	221
95.2	Business Task	221
95.3	SQL Solution	221
95.4	Explain It (B2)	221
95.5	Memory Hook	221
95.6	Common Mistake	221
96	Quick Drill	223
97	Question 4 — Orders Above £100	225
97.1	Recruiter Psychology	225
97.2	Business Task	225
97.3	SQL Solution	225
97.4	Explain It (B2)	225

97.5	Memory Hook	225
97.6	Common Mistake	225
98	Question 5 – Multiple Conditions	227
98.1	Recruiter Psychology	227
98.2	Business Task	227
98.3	SQL Solution	227
98.4	Explain It (B2)	227
98.5	Memory Hook	227
98.6	Common Mistake	227
99	Question 6 – OR Condition	229
99.1	Business Task	229
99.2	SQL Solution	229
99.3	Explain It (B2)	229
99.4	Memory Hook	229
100	Interview Tip	231
101	Question 7 – Sort Highest Revenue First	233
101.1	Recruiter Psychology	233
101.2	Business Task	233
101.3	SQL Solution	233
101.4	Explain It (B2)	233
101.5	Memory Hook	233
101.6	Common Mistake	233
102	Question 8 – Oldest Customers First	235
102.1	Business Task	235
102.2	SQL Solution	235
102.3	Explain It (B2)	235
103	Question 9 – Top Five Orders	237
103.1	Recruiter Psychology	237
103.2	Business Task	237
103.3	SQL Solution	237
103.4	Explain It (B2)	237
103.5	Memory Hook	237
104	Practice Challenge	239
105	Question 10 – Remove Duplicate Cities	241
105.1	Recruiter Psychology	241
105.2	Business Task	241
105.3	SQL Solution	241

105.4	Explain It (B2)	241
105.5	Memory Hook	241
105.6	Common Mistake	241
106	Question 11 — Find Products in Electronics	243
106.1	Business Task	243
106.2	SQL Solution	243
106.3	Explain It (B2)	243
107	Question 12 — Sort Products by Price	245
107.1	Business Task	245
107.2	SQL Solution	245
107.3	Explain It (B2)	245
108	SQL Interview Vocabulary	247
109	Mini Mock Interview	249
109.1	Interviewer	249
109.2	Interviewer	249
109.3	Interviewer	249
110	Cheat Sheet	251
110.1	SELECT	251
110.2	WHERE	251
110.3	ORDER BY	251
110.4	LIMIT	251
110.5	DISTINCT	251
111	Memory Hooks Summary	253
112	End-of-Sprint Challenge	255
113	SQL Bible	257
114	Sprint 2 — Aggregations	259
114.1	GROUP BY • HAVING • Aggregate Functions	259
115	Why Aggregations Matter	261
116	Memory Map	263
117	Aggregate Functions Overview	265
118	Question 13 — Count All Orders	267
118.1	Recruiter Psychology	267
118.2	Business Task	267
118.3	SQL Solution	267

118.4	Explain It (B2)	267
118.5	Memory Hook	267
118.6	Common Mistake	267
119	Question 14 — Count London Customers	269
119.1	Business Task	269
119.2	SQL Solution	269
119.3	Explain It (B2)	269
120	Question 15 — Total Revenue	271
120.1	Recruiter Psychology	271
120.2	Business Task	271
120.3	SQL Solution	271
120.4	Explain It (B2)	271
120.5	Memory Hook	271
121	Question 16 — Average Order Value	273
121.1	Business Task	273
121.2	SQL Solution	273
121.3	Explain It (B2)	273
121.4	Common Mistake	273
122	Question 17 — Largest Order	275
122.1	Business Task	275
122.2	SQL Solution	275
122.3	Explain It (B2)	275
123	Question 18 — Smallest Order	277
123.1	SQL Solution	277
123.2	Explain It (B2)	277
124	Understanding GROUP BY	279
125	Question 19 — Revenue by Customer	281
125.1	Recruiter Psychology	281
125.2	Business Task	281
125.3	SQL Solution	281
125.4	Explain It (B2)	281
125.5	Common Mistake	281
126	Question 20 — Orders by Status	283
126.1	Business Task	283
126.2	SQL Solution	283
126.3	Explain It (B2)	283

127	Question 21 — Customers by City	285
127.1	Business Task	285
127.2	SQL Solution	285
127.3	Explain It (B2)	285
128	GROUP BY Rule	287
129	Question 22 — Average Revenue by City	289
129.1	Business Task	289
129.2	SQL Solution	289
129.3	Explain It (B2)	289
130	HAVING	291
131	Question 23 — Cities with More Than Five Customers	293
131.1	Recruiter Psychology	293
131.2	SQL Solution	293
131.3	Explain It (B2)	293
132	Question 24 — Customers Spending Over £500	295
132.1	Business Task	295
132.2	SQL Solution	295
132.3	Explain It (B2)	295
133	WHERE vs HAVING	297
134	Question 25 — Top Revenue Cities	299
134.1	Business Task	299
134.2	SQL Solution	299
134.3	Explain It (B2)	299
135	Business Case	301
136	Mini Mock Interview	303
136.1	Interviewer	303
136.2	Interviewer	303
136.3	Interviewer	303
137	Cheat Sheet	305
137.1	COUNT	305
137.2	SUM	305
137.3	AVG	305
137.4	GROUP BY	305
137.5	HAVING	305
138	Memory Hooks	307

139	End-of-Sprint Challenge	309
140	SQL Bible	311
141	Sprint 3 — JOINS, CTEs & Subqueries	313
141.1	The Language of Relationships	313
142	The Evenec Retail Relationship Map	315
142.1	Entity Relationship Overview	315
143	Why JOIN Exists	317
144	JOIN Mental Model	319
145	Question 26 — INNER JOIN	321
145.1	Recruiter Psychology	321
145.2	Business Task	321
145.3	SQL Solution	321
145.4	Explain It (B2)	321
145.5	Memory Hook	321
146	INNER JOIN Visual	323
147	Question 27 — LEFT JOIN	325
147.1	Business Task	325
147.2	SQL	325
147.3	Explain It (B2)	325
147.4	Memory Hook	325
148	LEFT JOIN Visual	327
149	Interview Trap	329
150	Question 28 — Find Customers Without Orders	331
150.1	SQL	331
150.2	Explain It (B2)	331
151	Question 29 — Join Three Tables	333
152	JOIN Order	335
153	CTE — Common Table Expression	337
154	Question 30 — First CTE	339
154.1	Explain It (B2)	339
155	Why CTE Is Better	341

156	Subqueries	343
157	Question 31 — Above Average Orders	345
157.1	Explain It (B2)	345
158	Subquery Visual	347
159	Question 32 — Products Above Average Price	349
160	CTE vs Subquery	351
161	Business Case — Amazon Style	353
162	Mini Mock Interview	355
163	Cheat Sheet	357
163.1	INNER JOIN	357
163.2	LEFT JOIN	357
163.3	CTE	357
163.4	Subquery	357
164	Memory Hooks Summary	359
165	End-of-Sprint Challenge	361
166	SQL Bible	363
167	Sprint 4 — Window Functions & Interview Finale	365
167.1	ROW_NUMBER • RANK • DENSE_RANK • LAG • LEAD • OVER()	365
168	Why Window Functions Matter	367
169	The Window Mental Model	369
170	OVER()	371
171	Question 33 — ROW_NUMBER()	373
171.1	Recruiter Psychology	373
171.2	Business Task	373
171.3	Explain It (B2)	373
172	Visual	375
173	Question 34 — RANK()	377
174	Question 35 — DENSE_RANK()	379
175	Interview Trap	381

176	PARTITION BY	383
177	Visual	385
178	Question 36 — Top Customer in Every City	387
179	LAG()	389
180	LEAD()	391
181	Running Total	393
182	Moving Average	395
183	Window Function Cheat Sheet	397
184	Business Case 1 — Tesco	399
185	Business Case 2 — Amazon	401
186	Business Case 3 — Deliveroo	403
187	Business Case 4 — NHS	405
188	Business Case 5 — Wise	407
189	Business Case 6 — Monzo	409
190	Business Case 7 — Retail Dashboard	411
191	Business Case 8 — Customer Retention	413
192	Business Case 9 — Product Performance	415
193	Business Case 10 — Executive Report	417
194	Full Mock SQL Interview (45 Minutes)	419
194.1	Part 1 — Warm-up (10 min)	419
194.2	Part 2 — Aggregations (10 min)	419
194.3	Part 3 — JOINS (10 min)	419
194.4	Part 4 — Advanced SQL (15 min)	419
195	Interview English Cheat Sheet	421
196	Common Interview Mistakes	423
197	5-Minute Emergency Revision	425
197.1	SELECT	425
197.2	WHERE	425

197.3	GROUP BY	425
197.4	HAVING	425
197.5	JOIN	425
197.6	CTE	425
197.7	ROW_NUMBER	425
197.8	RANK	425
197.9	DENSE_RANK	425
197.10	LAG	425
197.11	LEAD	425
198	Final SQL Challenge	427
199	SQL Bible Complete	429

III

Part III — Excel Interview Bible

200	Excel Interview Bible	433
201	Sprint 1 — Excel Foundations	435
201.1	Tables • Filters • Sorting • Essential Navigation	435
202	The Evenec Retail Dataset	437
203	Why Companies Use Excel	439
204	Interview Question 1	441
204.1	Recruiter Psychology	441
204.2	Task	441
204.3	Steps	441
204.4	Why Tables Matter	441
204.5	Bogdan Notes	441
205	Interview Question 2	443
205.1	Filters	443
205.1.1	Steps	443
205.1.2	Interview English	443
206	Interview Question 3	445
206.1	Multi-Level Sorting	445
206.1.1	Steps	445
206.1.2	Why Recruiters Ask This	445
207	Freeze Panes	447
208	Interview Question 4	449
209	Essential Keyboard Shortcuts	451

210	Structured References	453
211	Interview Question 5	455
212	Removing Duplicates	457
213	Find & Replace	459
214	Data Validation	461
215	Conditional Formatting	463
216	Quick Quality Checks	465
217	Mini Mock Interview	467
218	Practice Challenge	469
219	Memory Hooks	471
220	Excel Interview Bible	473
221	Sprint 2 — XLOOKUP, INDEX/MATCH & Text Functions	475
221.1	Find • Clean • Combine • Validate	475
222	Why Lookup Functions Matter	477
223	Interview Question 6	479
223.1	XLOOKUP Basics	479
223.1.1	Recruiter Psychology	479
223.1.2	Business Task	479
223.1.3	Formula	479
223.1.4	Explain It (B2)	479
223.1.5	Memory Hook	479
223.1.6	Common Mistake	479
224	Visual	481
225	Why XLOOKUP Replaced VLOOKUP	483
226	Interview Question 7	485
226.1	Product Price Lookup	485
227	Missing Matches	487
228	INDEX + MATCH	489
229	XLOOKUP vs INDEX/MATCH	491
230	Interview Question 8	493
230.1	LEFT()	493

231	RIGHT()	495
232	MID()	497
233	LEN()	499
234	TRIM()	501
235	UPPER, LOWER, PROPER	503
236	TEXT()	505
237	CONCAT()	507
238	TEXTJOIN()	509
239	Cleaning Imported Data	511
240	Interview Question 9	513
241	Interview Question 10	515
242	Error Handling	517
243	Mini Business Case	519
244	Mini Mock Interview	521
245	Formula Cheat Sheet	523
246	Memory Hooks	525
247	End-of-Sprint Challenge	527
248	Excel Interview Bible	529
249	Sprint 3 — Manager's Monday Morning Dashboard	531
249.1	Pivot Tables • Pivot Charts • Slicers • Timeline • Power Query	531
250	What We'll Build	533
251	Dashboard Preview	535
252	Download the Real Dashboard	537
253	Step 1 — Import Data with Power Query	539
253.1	Recruiter Psychology	539
253.2	Business Task	539
253.2.1	Steps	539
253.2.2	Explain It (B2)	539

254	Why Power Query Matters	541
255	Step 2 — Build Your First Pivot Table	543
255.1	Business Task	543
255.1.1	Steps	543
255.1.2	Result	543
255.1.3	Explain It (B2)	543
256	Question 11	545
257	Step 3 — Sort Highest Revenue	547
258	Step 4 — Create a Pivot Chart	549
259	Chart Psychology	551
260	Step 5 — Revenue by Category	553
261	Step 6 — Add Slicers	555
262	Step 7 — Add Timeline	557
263	Step 8 — Build KPI Cards	559
264	KPI Layout	561
265	Formatting Like a Professional	563
266	Business Case — Monday Morning	565
267	Common Dashboard Mistakes	567
268	Mini Mock Interview	569
269	Dashboard Checklist	571
270	Power Query Cheat Sheet	573
271	End-of-Sprint Challenge	575
272	Excel Interview Bible	577
273	Sprint 4 — Excel Assessment Centre	579
273.1	Dashboards • Business Cases • Mock Interview • 30-Minute Challenge . . .	579
274	Assessment Overview	581
275	Before You Start	583
276	Assessment Task	585

277	Task 1 – Revenue KPI	587
278	Task 2 – Total Orders	589
279	Task 3 – Average Order Value	591
280	KPI Layout	593
281	Task 4 – Revenue by City	595
282	Task 5 – Best Selling Category	597
283	Task 6 – Interactive Dashboard	599
284	Dashboard Design Rules	601
285	Business Case 1 – CEO Report	603
286	Business Case 2 – Operations	605
287	Business Case 3 – Customer Growth	607
288	Business Case 4 – Executive Summary	609
289	Interview Question 11	611
290	Interview Question 12	613
291	Interview Question 13	615
292	Interview Question 14	617
293	Mini Mock Interview	619
294	The 30-Minute Assessment	621
295	Rebuild the Dashboard Yourself	623
295.0.1	Your Mission	623
296	Assessment Rubric	625
297	Recruiter Red Flags	627
298	B2 Presentation Script	629
299	Emergency Interview Cheat Sheet	631
300	Final Excel Challenge	633
301	Excel Bible Complete	635

IV	Part IV — Power BI Interview Bible	
302	Power BI Interview Bible	639
303	Sprint 1 — Power BI Foundations	641
303.1	Import • Data Model • Power Query • First Visuals	641
304	Why Companies Use Power BI	643
305	Interview Question 1	645
305.1	Why Power BI?	645
306	Step 1 — Create Your First Report	647
307	Power Query Opens Automatically	649
308	Interview Question 2	651
308.1	Power Query Basics	651
309	Close & Apply	653
310	Step 2 — Build the Data Model	655
311	Star Schema	657
312	Interview Question 3	659
313	Step 3 — Create Your First Visual	661
314	Visual Types	663
315	Step 4 — Create KPI Cards	665
316	Dashboard Layout	667
317	Step 5 — Add Filters	669
318	Formatting Like a Professional	671
319	Mini Business Case	673
320	Mini Mock Interview	675
321	Power BI Cheat Sheet	677
322	End-of-Sprint Challenge	679
323	Power BI Interview Bible	681

324	Sprint 2 — Thinking in DAX	683
324.1	Measures • CALCULATE • FILTER • RELATED • ALL	683
325	What Makes DAX Different?	685
326	Understanding Context	687
327	Two Types of Context	689
328	Interview Question 4	691
329	Your First Measure	693
330	Calculated Column vs Measure	695
331	Interview Question 5	697
332	COUNT vs COUNTROWS	699
333	AVERAGE	701
334	CALCULATE	703
335	Why CALCULATE Matters	705
336	FILTER	707
337	RELATED	709
338	RELATEDTABLE	711
339	ALL	713
340	DIVIDE	715
341	Time Intelligence Preview	717
342	Mini Business Case	719
343	Mini Business Case	721
344	Common DAX Mistakes	723
345	DAX Cheat Sheet	725
346	Mini Mock Interview	727
347	End-of-Sprint Challenge	729
348	Memory Hooks	731

349	Power BI Interview Bible	733
350	Sprint 3 — Executive Dashboard	735
350.1	KPI Cards • Slicers • Drill-through • Tooltips • Bookmarks	735
351	What We'll Build	737
352	Dashboard Preview	739
353	Dashboard Design Principles	741
354	Step 1 — Build KPI Cards	743
355	Top City Measure	745
356	Interview Question 6	747
357	Step 2 — Revenue by City	749
358	Step 3 — Revenue by Category	751
359	Step 4 — Date Trend	753
360	Step 5 — Slicers	755
361	Step 6 — Sync Slicers	757
362	Step 7 — Drill-through	759
363	Step 8 — Custom Tooltips	761
364	Step 9 — Bookmarks	763
365	Navigation Buttons	765
366	Conditional Formatting	767
367	Business Case	769
368	Storytelling with Data	771
369	Mini Mock Interview	773
370	Executive Dashboard Checklist	775
371	Power BI Dashboard Cheat Sheet	777
372	End-of-Sprint Challenge	779
373	Presentation Script (B2)	781

374	Memory Hooks	783
375	Power BI Interview Bible	785
376	Sprint 4 — Power BI Assessment Centre	787
376.1	45-Minute Challenge • Executive Presentation • Mock Interview	787
377	Assessment Overview	789
378	Before You Start	791
379	Assessment Scenario	793
380	Your Mission	795
381	Dashboard Blueprint	797
382	The 45-Minute Timer	799
383	Assessment Task 1	801
383.1	Revenue Overview	801
384	Assessment Task 2	803
384.1	City Performance	803
385	Assessment Task 3	805
385.1	Category Analysis	805
386	Assessment Task 4	807
386.1	Monthly Trend	807
387	Assessment Task 5	809
387.1	Interactive Experience	809
388	Assessment Task 6	811
388.1	Drill-through Page	811
389	Assessment Task 7	813
389.1	Custom Tooltips	813
390	Assessment Task 8	815
390.1	Navigation	815
391	Hiring Manager Questions	817
391.1	Question 1	817
391.2	Question 2	817
391.3	Question 3	817
391.4	Question 4	817

392	Recruiter Questions	819
392.1	Tell me about this dashboard.	819
392.2	What would you improve next?	819
393	Presentation Challenge	821
394	Assessment Rubric	823
395	Recruiter Red Flags	825
396	Final Checklist	827
397	Companion Files	829
398	Final Interview Mindset	831
399	Congratulations	833
400		835
401		837

1. evenec

2. Data Analyst Interview Bible (UK 2026)

2.1 Part 1 — HR Interview Mastery

First Edition

Authors

- Bohdan Shevchuk
- ChatGPT (OpenAI)

2.2 Part 2 — SQL Bible

This section prepares you for SQL interviews in UK companies using one consistent business dataset:

Evenec Retail.

Topics include:

- SELECT & WHERE
- GROUP BY & HAVING
- JOINS
- CTEs
- Window Functions
- Business Cases
- Full Mock Interview

CC BY 4.0

3.

4.

I

74	Question 45 — How do you stay organised when learning new skills?	177
74.1	Recruiter Psychology	177
74.2	Model Answer	177
75	Question 46 — Tell me about making a decision with limited information.	179
75.1	Recruiter Psychology	179
75.2	Model Answer	179
76	Practice Break	181
77	Question 47 — What motivates you to keep learning?	183
77.1	Recruiter Psychology	183
77.2	Model Answer	183
78	Question 48 — Describe a time you worked carefully with details.	185
78.1	Recruiter Psychology	185
78.2	Model Answer	185
79	Question 49 — What would your previous colleagues say about you? .	187
79.1	Recruiter Psychology	187
79.2	Model Answer	187
80	Question 50 — Is there anything else you'd like us to know?	189
80.1	Recruiter Psychology	189
80.2	Model Answer	189
81	Chapter Summary	191
81.1	Five Final Memory Hooks	191
82	Salary Negotiation (UK £35–45k) . .	193
82.1	When should you discuss salary?	193
82.2	If HR asks first	193
82.3	If they offer £32,000	193
82.4	If they ask for your minimum salary	193
82.5	Salary Negotiation Checklist	194
83	Recruiter Red Flags	195
83.1	Green Flags	195
83.2	Yellow Flags	195
83.3	Red Flags	195
84	Mini Mock Interview	197
84.1	Interview Starts	197
85	Self-Evaluation	199
86	Final Interview Checklist	201
86.1	The Day Before	201
86.2	One Hour Before	201
86.3	During the Interview	201
86.4	After the Interview	201
87	Part 1 Complete	203

5. HR Interview Mastery

6. Questions 1–15

This chapter covers the first fifteen HR interview questions most commonly asked during Junior Data Analyst interviews in the United Kingdom.

The answers are written in **natural B2 English** and are designed to sound confident, professional and conversational.

7. How to Use This Chapter

Each question follows the same structure.

- **Question** – what the interviewer asks.
 - **Recruiter Psychology** – what they are really evaluating.
 - **Model Answer** – natural B2 English.
 - **Bogdan Note** – how to adapt it to your real experience.
 - **Memory Hook** – a quick phrase for remembering the structure.
 - **Follow-up Questions** – what usually comes next.
 - **Common Mistake** – what to avoid.
-

8. Question 1 — Tell me about yourself

This is almost always the first question.

The interviewer is not asking for your life story.

They are checking whether you can introduce yourself clearly.

8.1 Recruiter Psychology

They evaluate:

- structured thinking;
- communication;
- confidence;
- career direction.

8.2 Model Answer

I'm currently working in a dental laboratory in London as a 3D Designer, where I work with digital workflows and attention to detail every day.

Before this role, I ran my own business in Ukraine, where I regularly worked with sales reports and customer information to make business decisions.

That experience made me interested in data analysis, so I completed the GoIT Data Analytics programme and learned SQL, Excel, Power BI and Python.

Now I'm looking for my first Junior Data Analyst role where I can combine business understanding with analytical skills.

Bogdan Note

Use your real transition story.

Business → Dental Lab → GoIT → Data Analyst.

This makes your career change feel intentional.

Memory Hook

Past → Present → Future

8.2.1 Follow-up Questions

- Why did you change careers?
- What did you enjoy about your business?
- Why data analytics?

8.2.2 Common Mistake

Don't say:

"I love numbers."

Instead, connect your answer to real experience.

9. Question 2 — Why do you want to become a Data Analyst?

9.1 Recruiter Psychology

The interviewer wants to know whether your motivation is sustainable.

They're checking if this is a thoughtful career move rather than a temporary idea.

9.2 Model Answer

I enjoy solving business problems with data.

In my previous business I often used sales reports and customer information to make decisions.

I realised I wanted stronger technical skills, so I completed a Data Analytics programme.

Now I want to use both my business experience and analytical skills in a professional team.

💡 Bogdan Note

Mention business decisions instead of saying you've always loved data.

i Memory Hook

Business → Learning → Career

9.2.1 Common Mistake

Don't say:

"Because analysts earn good salaries."

10. Question 3 — Why are you leaving your current job?

10.1 Recruiter Psychology

The interviewer wants to see whether you leave professionally.
They're checking for maturity rather than complaints.

10.2 Model Answer

I've learned valuable skills in my current role, especially attention to detail and digital workflows.
However, my long-term goal is to build my career in data analytics.
I'm looking for a role where I can grow professionally and use the skills I've been developing through my training.

i Memory Hook

Positive → Growth → Future

10.2.1 Follow-up Questions

- What did you learn?
- What will you miss?

10.2.2 Common Mistake

Never criticise your current employer.

11. Practice Break

Answer these aloud.

1. Tell me about yourself.
2. Why Data Analytics?
3. Why are you leaving your current job?

Target time:

45–60 seconds each.

12. Question 4 — Why do you want to work for our company?

12.1 Recruiter Psychology

This question checks whether you prepared.

A candidate who researches the company usually appears more motivated.

12.2 Model Answer

I like that your company makes decisions based on data and invests in digital improvement.

I also appreciate the opportunity to learn from experienced analysts and work on real business problems.

I'm looking for a place where I can grow while contributing from the beginning.

Bogdan Note

Before every interview, spend five minutes checking the company's website, LinkedIn page and recent news.

12.2.1 Company Research Checklist

- Website
- LinkedIn
- News
- Products
- Values

12.2.2 Common Mistake

Never answer:

"I just need a job."

13. Question 5 — What are your strengths?

13.1 Recruiter Psychology

The interviewer wants strengths supported by evidence.

13.2 Model Answer

I'd highlight three strengths.

First, attention to detail.

Second, problem-solving.

Third, business understanding from running my own company.

 Bogdan Note

Always attach a real example to each strength.

13.2.1 Memory Hook

Three strengths.

13.2.2 Common Mistake

Listing ten strengths without examples.

14. Question 6 — What is your biggest weakness?

14.1 Recruiter Psychology

This isn't a trap.

They want to see self-awareness.

14.2 Model Answer

Earlier I sometimes spent too much time trying to make every detail perfect.

I've learned to balance quality with deadlines by prioritising the most important tasks first.

14.2.1 Good Weaknesses

- perfectionism
- asking for help too late
- public speaking while improving

14.2.2 Common Mistake

Don't use fake weaknesses like:

"I work too hard."

15. Practice Break

Ask yourself:

- Which strength has the strongest real example?
 - Which weakness shows genuine improvement?
-

16. Question 7 — Tell me about a challenge you faced.

16.1 Recruiter Psychology

This question measures problem-solving.
Use STAR.

16.2 Model Answer

16.2.1 Situation

Our workflow had delays.

16.2.2 Task

Reduce processing time.

16.2.3 Action

Reviewed the workflow and improved communication.

16.2.4 Result

The process became smoother and deadlines became easier to meet.

i Memory Hook

STAR
Situation → Task → Action → Result

16.2.5 Bogdan Note

Later we'll replace this with your strongest real business example.

17. Question 8 — Tell me about a mistake you made.

17.1 Recruiter Psychology

Recruiters don't expect perfection.
They expect accountability.

17.2 Model Answer

In one project I focused too much on one part of the work before checking the wider process.
I realised this created unnecessary delays.
Since then I've started reviewing priorities earlier and communicating progress more regularly.

17.2.1 Common Mistake

Never blame other people. _____

18. Question 9 — How do you prioritise your work?

18.1 Recruiter Psychology

Analysts constantly balance multiple requests.

The interviewer wants to understand your decision process.

18.2 Model Answer

I usually start by identifying deadlines and business impact.

Then I break larger tasks into smaller steps and review priorities during the day if something changes.

i Memory Hook

Impact → Deadline → Dependencies

18.2.1 Common Mistake

Saying:

“I just do everything.”

19. Practice Break

Answer Questions 7–9 in under **90 seconds** each.
Focus on speaking naturally rather than perfectly.

20. Question 10 — Tell me about working with difficult people.

20.1 Recruiter Psychology

This question tests emotional intelligence.

20.2 Model Answer

I try to understand their perspective first.

Then I focus on solving the problem instead of making it personal.

Clear communication usually helps avoid misunderstandings.

20.2.1 Bogdan Note

Use a real workplace example without criticising colleagues.

21. Question 11 — Describe teamwork.

21.1 Recruiter Psychology

Junior analysts rarely work alone.
They're checking collaboration.

21.2 Model Answer

I enjoy working in teams because different people bring different strengths.
In my current role I regularly communicate with colleagues to make sure work moves smoothly between different stages.

21.2.1 Memory Hook

Communication → Reliability → Support.

22. Question 12 — How do you handle pressure?

22.1 Recruiter Psychology

The interviewer wants confidence under deadlines.

22.2 Model Answer

I stay calm by breaking large tasks into smaller actions.

I also keep track of priorities so I focus on the most important work first.

i Memory Hook

Break → Prioritise → Execute

22.2.1 Common Mistake

Don't pretend pressure never affects you.

23. Question 13 — Where do you see yourself in five years?

23.1 Recruiter Psychology

They're checking ambition without unrealistic expectations.

23.2 Model Answer

In five years I'd like to become a confident Data Analyst with strong technical skills and experience solving real business problems.

I'd also like to mentor newer colleagues as I continue developing professionally.

23.2.1 Common Mistake

Don't say:

"I want your manager's job."

24. Question 14 — Why should we hire you?

24.1 Recruiter Psychology

This is your value statement.

24.2 Model Answer

I bring a combination of business experience, attention to detail and technical skills.

I've already invested time in learning SQL, Excel, Power BI and Python, and I'm motivated to continue growing while contributing to the team.

 Bogdan Note

Keep this answer around 30 seconds.

24.2.1 Memory Hook

Business + Technical + Motivation.

25. Question 15 — Do you have any questions for us?

25.1 Recruiter Psychology

This is often the final impression.
Good questions show curiosity.

25.1.1 Model Answer

I'd love to know what success looks like during the first three months in this role.
I'd also be interested in learning about the team's projects and development opportunities.

25.1.2 Five Excellent Questions

- What does success look like in the first three months?
- What projects would I work on first?
- How does the team collaborate?
- What learning opportunities do junior analysts receive?
- Which tools does the team use most often?

25.1.3 Common Mistake

Never answer:

"No, I don't have any questions."

26. Chapter Summary

26.1 Top Five Memory Hooks

Concept	Hook
Self-introduction	Past → Present → Future
STAR	Situation → Task → Action → Result
Prioritisation	Impact → Deadline → Dependencies
Pressure	Break → Prioritise → Execute
Company Research	Website → LinkedIn → News

27. 5-Minute Interview Drill

Set a timer.

Answer these five questions without reading your notes.

Question	Target
Tell me about yourself	60 s
Why Data Analytics?	45 s
Biggest strength	30 s
Biggest weakness	45 s
Why this company?	45 s

Listen to your recording afterwards.

Improve **one sentence**, not the whole answer.

Small improvements repeated consistently create confident interviews. {callout-tip title="Practice Before Your Interview"}

Before moving to the next chapter:

- Answer each question aloud in English.
- Keep each answer between **60–90 seconds**.
- Use the **STAR** framework whenever possible.
- Record yourself and listen for clarity and confidence.

The goal is not to memorise answers, but to communicate naturally.

28. HR Interview Mastery

29. Questions 16–30

This chapter continues the HR interview preparation with behavioural and workplace questions that frequently appear during Junior Data Analyst interviews in the UK.

The goal is not to memorise perfect sentences but to understand how recruiters evaluate your thinking and communication.

30. Question 16 — Tell me about a time you learned something new quickly.

30.1 Recruiter Psychology

Recruiters want evidence that you can learn new tools and adapt to change.

30.2 Model Answer

When I decided to move into data analytics, I needed to learn several new tools in a relatively short time.

I completed the GoIT Data Analytics programme, practised SQL, Excel, Power BI and Python, and built projects to apply what I learned.

That experience taught me how to organise my learning and stay consistent.

 Bogdan Note

This is one of your strongest career-transition stories.

 Memory Hook

Learn → Practice → Apply

30.2.1 Common Mistake

Don't simply say:

"I'm a fast learner."

Show evidence.

31. Question 17 — Describe a time you improved a process.

31.1 Recruiter Psychology

Analysts improve processes constantly.

Recruiters want initiative.

31.2 Model Answer

In my previous work I reviewed how tasks moved through our workflow.

I noticed unnecessary delays between steps, suggested a clearer process and improved communication.

The workflow became more organised and easier to manage.

31.2.1 Memory Hook

Observe → Improve → Result.

32. Question 18 — Tell me about a time you worked with data.

32.1 Recruiter Psychology

Even if this isn't a formal analyst role, recruiters want transferable experience.

32.2 Model Answer

In my business I regularly reviewed sales information and customer data to understand performance.
I used that information when making decisions about products and priorities.
That experience became one of the reasons I became interested in data analytics.

 Bogdan Note

This answer connects your business experience directly to analytics.

33. Practice Break

Answer Questions 16–18 without reading.

Target:

- 60 seconds each.

34. Question 19 — How do you deal with repetitive tasks?

34.1 Recruiter Psychology

Data work often includes repetitive processes.
The interviewer wants discipline.

34.2 Model Answer

I try to stay focused by understanding why the task matters.
I also look for opportunities to organise my work more efficiently while maintaining accuracy.

34.2.1 Memory Hook

Purpose → Accuracy → Efficiency.

34.2.2 Common Mistake

Don't say repetitive work is boring.

35. Question 20 — Tell me about receiving feedback.

35.1 Recruiter Psychology

Can you accept feedback professionally?

35.2 Model Answer

I see feedback as an opportunity to improve.

In previous roles I've received suggestions that helped me become more organised and communicate more clearly.

I try to apply feedback as quickly as possible.

35.2.1 Memory Hook

Listen → Apply → Improve.

36. Question 21 — Tell me about giving feedback.

36.1 Recruiter Psychology

This measures communication and professionalism.

36.2 Model Answer

I try to give feedback respectfully and focus on the work rather than the person.
I usually explain what worked well first before suggesting improvements.

36.2.1 Common Mistake

Avoid sounding confrontational.

37. Question 22 — How do you organise your work?

37.1 Recruiter Psychology

Organisation is essential for analysts.

37.2 Model Answer

I usually start by identifying priorities and deadlines.

Then I create a simple task list and review my progress during the day.

This helps me stay organised without losing focus.

i Memory Hook

Plan → Track → Review.

38. Practice Break

Ask yourself:

- Which answer sounds the most natural?
 - Which example feels strongest?
-

39. Question 23 — Tell me about a time you solved a problem.

39.1 Recruiter Psychology

Problem-solving is one of the core skills for analysts.

39.2 Model Answer

I once noticed that a workflow was creating unnecessary delays.

I reviewed the process, identified where communication was slowing things down and suggested a clearer approach.

The workflow became more efficient afterwards.

39.2.1 Memory Hook

Problem → Analysis → Solution → Result.

40. Question 24 — How do you stay motivated?

40.1 Recruiter Psychology

Recruiters want consistent motivation.

40.2 Model Answer

I stay motivated when I can see progress.

I enjoy learning new skills and solving practical problems, so I like setting small goals and improving step by step.

40.2.1 Common Mistake

Don't rely only on external motivation.

41. Question 25 — What does success mean to you?

41.1 Recruiter Psychology

The interviewer wants realistic professional values.

41.2 Model Answer

For me, success means delivering reliable work, continuing to learn and helping the team achieve good results.

I also think success includes building skills that allow me to take on greater responsibility over time.

41.2.1 Memory Hook

Reliable → Learn → Grow.

42. Practice Break

Record your answers to Questions 23–25.

Listen for:

- clarity
 - pace
 - confidence
-

43. Question 26 — Tell me about working under a deadline.

43.1 Recruiter Psychology

Deadlines are part of everyday analytical work.

43.2 Model Answer

When I work under deadlines, I first identify the highest-priority tasks.
Then I break the work into manageable steps and monitor my progress.
This helps me stay calm and deliver quality work.

43.2.1 Memory Hook

Prioritise → Break Down → Deliver.

44. Question 27 — Describe a situation where you took initiative.

44.1 Recruiter Psychology

Recruiters want proactive people.

44.2 Model Answer

I noticed an opportunity to improve part of our workflow without being asked.

I suggested a practical change, discussed it with the team and helped implement it.

The process became smoother afterwards.

 Bogdan Note

Use examples where you solved problems instead of waiting for instructions.

45. Question 28 — How do you deal with uncertainty?

45.1 Recruiter Psychology

Analysts often work with incomplete information.

45.2 Model Answer

I try to gather as much relevant information as possible before making a decision.

If something is unclear, I ask questions instead of making assumptions.

I believe good communication reduces uncertainty.

45.2.1 Memory Hook

Gather → Ask → Decide.

46. Question 29 — Describe a time you had multiple priorities.

46.1 Recruiter Psychology

This tests workload management.

46.2 Model Answer

I've had situations where several important tasks needed attention at the same time.

I reviewed deadlines, considered business impact and completed the highest-priority work first.

That helped me meet expectations without losing quality.

46.2.1 Common Mistake

Don't say you simply worked longer hours.

47. Question 30 — Why should we believe you'll succeed as a Junior Data Analyst without previous analyst experience?

47.1 Recruiter Psychology

This is one of the most important transition questions.
Recruiters want confidence supported by evidence.

47.2 Model Answer

Although this would be my first professional Data Analyst role, I already have experience making business decisions using information, working carefully with digital workflows and learning new technical skills.

I've invested significant time in SQL, Excel, Power BI and Python, and I'm ready to continue learning while contributing to the team from day one.

Bogdan Note

This is your strongest closing answer for career-transition interviews.

Memory Hook

Experience + Skills + Motivation.

47.2.1 Common Mistake

Never apologise for being junior.
Instead, emphasise what you already bring.

48. Chapter Summary

48.1 Five Key Memory Hooks

Concept	Memory Hook
Learning	Learn → Practice → Apply
Organisation	Plan → Track → Review
Problem Solving	Problem → Analysis → Solution
Deadlines	Prioritise → Break Down → Deliver
Career Transition	Experience + Skills + Motivation

49. 5-Minute Interview Drill

Answer these questions without looking at your notes.

Question	Target
Learned something quickly	60 s
Improved a process	60 s
Worked with data	45 s
Solved a problem	60 s
Why you'll succeed as a Junior Data Analyst	60 s

Focus on speaking naturally rather than memorising every word.

Your goal is confidence, not perfection. {callout-tip title="Practice Before Your Interview"}
Before moving to the next chapter:

- Answer each question aloud in English.
- Keep each answer between **60–90 seconds**.
- Use the **STAR** framework whenever possible.
- Record yourself and listen for clarity and confidence.

The goal is not to memorise answers, but to communicate naturally.

50. HR Interview Mastery

51. Questions 31–40

This chapter focuses on ownership, communication, adaptability and professional judgement.

These questions are common in behavioural interviews because recruiters want to understand how you think when situations become less predictable.

52. Question 31 — Tell me about a time you had to learn from failure.

52.1 Recruiter Psychology

The interviewer wants to see resilience.

They're looking for someone who reflects, learns and improves instead of becoming defensive.

52.2 Model Answer

One experience taught me that focusing only on immediate tasks without reviewing the bigger picture can create unnecessary delays.

I reflected on what happened, changed how I planned my work and started checking priorities earlier.

Since then I've become more organised when managing multiple responsibilities.

Bogdan Note

Focus on the lesson rather than the failure itself.

Memory Hook

Reflect → Adjust → Improve.

52.2.1 Common Mistake

Don't choose a failure that makes you sound careless or irresponsible.

53. Question 32 — Describe a time you had to explain something complicated.

53.1 Recruiter Psychology

Analysts often explain technical information to non-technical people.
Communication matters as much as technical knowledge.

53.2 Model Answer

I try to explain complex ideas using simple language and practical examples.
I usually avoid unnecessary technical terms and check whether the other person understands before continuing.
Clear communication helps everyone make better decisions.

53.2.1 Memory Hook

Simple → Example → Confirm.

53.2.2 Follow-up Questions

- How do you know someone understands?
 - Can you give an example?
-

54. Question 33 — How do you handle changing priorities?

54.1 Recruiter Psychology

Business priorities change constantly.
They're checking flexibility.

54.2 Model Answer

When priorities change, I first understand what has become most important for the business.
Then I reorganise my tasks and communicate if timelines need to change.
Staying flexible helps me deliver the most valuable work first.

i Memory Hook

Understand → Reorganise → Communicate.

54.2.1 Common Mistake

Don't say changing priorities frustrates you.

55. Practice Break

Answer Questions 31–33.

Focus on speaking naturally.

Target:

- 60 seconds each.
-

56. Question 34 — Tell me about a time you had to make a difficult decision.

56.1 Recruiter Psychology

They're evaluating judgement.

56.2 Model Answer

I've had situations where I needed to balance several competing priorities.

I gathered the available information, considered the potential impact and made the decision that best supported the overall goal.

Afterwards I reviewed the outcome so I could learn from the experience.

56.2.1 Memory Hook

Gather → Decide → Review.

57. Question 35 — What would you do if you didn't know the answer?

57.1 Recruiter Psychology

This is an honesty test.

Good analysts don't pretend to know everything.

57.2 Model Answer

If I didn't know the answer, I'd first research the problem using available resources.

If I still needed help, I'd ask a colleague while explaining what I'd already tried.

I believe learning efficiently is more important than pretending to know everything.

 Bogdan Note

This answer shows independence before asking for help.

57.2.1 Common Mistake

Never say you'd guess. _____

58. Question 36 — Tell me about a time you worked independently.

58.1 Recruiter Psychology

Junior analysts need supervision, but they also need initiative.

58.2 Model Answer

In previous roles I've managed responsibilities independently by planning my work and tracking my progress.

When necessary, I communicated updates and asked questions early rather than waiting until problems became bigger.

That helped me stay responsible while working efficiently.

58.2.1 Memory Hook

Plan → Execute → Update.

59. Practice Break

Record yourself answering Questions 34–36.

Listen for:

- confidence
 - clarity
 - pace
-

60. Question 37 — Describe a situation where you had to be very accurate.

60.1 Recruiter Psychology

Accuracy is one of the most important qualities for analysts.

60.2 Model Answer

In my current work, accuracy is essential because small mistakes can affect later stages of the process. I developed habits like double-checking important details and following consistent workflows. Those habits are directly relevant to analytical work.

 Bogdan Note

This is a great opportunity to reference your dental laboratory experience.

 Memory Hook

Check → Verify → Deliver.

60.2.1 Common Mistake

Don't simply say you're detail-oriented.
Explain how you ensure accuracy.

61. Question 38 — How do you react when someone disagrees with you?

61.1 Recruiter Psychology

Conflict management matters.
They're looking for professionalism.

61.2 Model Answer

I try to understand why the other person sees the situation differently.
Then I focus on discussing facts rather than personalities.
Even if we disagree, I believe respectful communication usually leads to better decisions.

61.2.1 Memory Hook

Listen → Understand → Discuss Facts.

62. Question 39 — Tell me about a goal you achieved.

62.1 Recruiter Psychology

They want evidence that you finish what you start.

62.2 Model Answer

One goal I achieved was successfully completing my transition toward data analytics.

I balanced work with learning SQL, Excel, Power BI and Python while completing practical projects.

That experience showed me the importance of consistency over time.

 Bogdan Note

Use your career transition as a personal achievement.

62.2.1 Common Mistake

Choose a goal with a measurable result whenever possible.

63. Question 40 — Why should we choose you over another candidate?

63.1 Recruiter Psychology

This is your positioning statement.

They're asking what makes you different.

63.2 Model Answer

I believe I offer a valuable combination of business experience, technical learning and a strong willingness to grow.

I've already demonstrated that I can learn new skills, adapt to change and work carefully with detailed processes.

I'm ready to bring that mindset into my first Junior Data Analyst role.

i Memory Hook

Business + Learning + Reliability.

63.2.1 Common Mistake

Don't compare yourself negatively with other candidates.

Focus on your own value.

64. Chapter Summary

64.1 Five Key Memory Hooks

Concept	Memory Hook
Learning from failure	Reflect → Adjust → Improve
Explaining complexity	Simple → Example → Confirm
Changing priorities	Understand → Reorganise → Communicate
Decision making	Gather → Decide → Review
Accuracy	Check → Verify → Deliver

65. STAR Practice

Complete these prompts using the STAR framework.

Prompt	Time
Learned from failure	90 s
Explained something complicated	90 s
Worked independently	90 s
Achieved a goal	90 s

Remember:

- Situation
 - Task
 - Action
 - Result
-

66. End-of-Chapter Reflection

Ask yourself these questions.

- Which answer sounds most natural?
- Which story feels strongest?
- Which answer still sounds memorised?
- Which example could become more specific?

Your goal is not perfect English.

Your goal is sounding like a confident future colleague. {callout-tip title="Practice Before Your Interview"}

Before moving to the next chapter:

- Answer each question aloud in English.
- Keep each answer between **60–90 seconds**.
- Use the **STAR** framework whenever possible.
- Record yourself and listen for clarity and confidence.

The goal is not to memorise answers, but to communicate naturally.

67. HR Interview Mastery

68. Questions 41–50

This final chapter completes the HR Interview Mastery section with advanced behavioural questions, salary negotiation strategies for the UK market, recruiter red flags, a realistic mock interview and practical checklists.

The goal is to help you finish an interview with confidence.

69. Question 41 — Tell me about a time you disagreed with a decision.

69.1 Recruiter Psychology

They're checking whether you can disagree professionally.

69.2 Model Answer

I try to understand the reasoning behind a decision before reacting.

If I have concerns, I explain them respectfully and focus on facts rather than emotions.

Even when a different decision is made, I continue supporting the team's goal.

i Memory Hook

Listen → Explain → Support.

69.2.1 Common Mistake

Don't describe yourself as someone who always argues.

70. Question 42 — Tell me about adapting to change.

70.1 Recruiter Psychology

Analysts work in changing environments.

70.2 Model Answer

I've experienced several major changes during my career, including moving to London and changing careers into data analytics.

Those experiences taught me how to learn quickly and stay flexible while continuing to work toward long-term goals.

 Bogdan Note

Your move to London is a strong real-life example of adaptability.

70.2.1 Memory Hook

Adapt → Learn → Grow. _____

71. Question 43 — How do you handle constructive criticism?

71.1 Recruiter Psychology

They're evaluating coachability.

71.2 Model Answer

I appreciate constructive feedback because it helps me improve.

I try to understand the suggestion, apply it in my work and review whether the change produces better results.

71.2.1 Memory Hook

Listen → Apply → Improve.

72. Practice Break

Answer Questions 41–43.

Target:

- 45–60 seconds each.
-

73. Question 44 — Describe a situation where you helped someone.

73.1 Recruiter Psychology

They're checking collaboration.

73.2 Model Answer

I enjoy helping colleagues when I can.

Sometimes sharing knowledge or explaining a process saves time for the whole team.

I believe teamwork becomes stronger when people support each other.

73.2.1 Common Mistake

Don't make yourself sound like the only person who solves problems.

74. Question 45 — How do you stay organised when learning new skills?

74.1 Recruiter Psychology

Junior analysts learn constantly.

74.2 Model Answer

I usually divide large topics into smaller learning goals.

Then I practise regularly and apply what I've learned through real exercises or projects.

That approach helped me learn SQL, Excel, Power BI and Python.

i Memory Hook

Small Goals → Practice → Projects.

75. Question 46 — Tell me about making a decision with limited information.

75.1 Recruiter Psychology

Real business decisions rarely have perfect information.

75.2 Model Answer

When information is limited, I collect the most relevant facts available and avoid making unnecessary assumptions.

If needed, I ask questions to reduce uncertainty before deciding.

My goal is making the most informed decision possible.

75.2.1 Memory Hook

Gather → Clarify → Decide.

76. Practice Break

Ask yourself:

- Which answer feels most natural?
 - Which example sounds strongest?
-

77. Question 47 — What motivates you to keep learning?

77.1 Recruiter Psychology

They want long-term learners.

77.2 Model Answer

I enjoy seeing measurable progress.

Learning new skills becomes more motivating when I can immediately apply them to practical problems.

That's one reason I enjoy data analytics.

77.2.1 Memory Hook

Progress → Application → Motivation.

78. Question 48 — Describe a time you worked carefully with details.

78.1 Recruiter Psychology

Accuracy is critical.

78.2 Model Answer

In my current work, small details can affect later stages of production.
I developed habits like checking important details twice and following consistent workflows.
Those habits are directly relevant to analytical work.

 Bogdan Note

Your dental laboratory experience is one of your strongest proof points.

78.2.1 Memory Hook

Check → Verify → Deliver. _____

79. Question 49 — What would your previous colleagues say about you?

79.1 Recruiter Psychology

They're checking self-awareness.

79.2 Model Answer

I think they'd describe me as reliable, organised and willing to help.

I'd also hope they'd say I'm someone who learns quickly and stays calm when solving problems.

79.2.1 Common Mistake

Don't exaggerate.

Keep it believable.

80. Question 50 — Is there anything else you'd like us to know?

80.1 Recruiter Psychology

This is your final opportunity.
Finish confidently.

80.2 Model Answer

I'd like you to know that I'm genuinely committed to building my career in data analytics.
I've already invested significant time in developing technical skills while gaining valuable experience in business and digital workflows.
I'm excited about contributing to a team where I can continue learning and creating value.

i Memory Hook

Commitment → Skills → Value.

80.2.1 Final Impression Tip

End with confidence.
Pause.
Smile.

81. Chapter Summary

81.1 Five Final Memory Hooks

Concept	Memory Hook
Disagreement	Listen → Explain → Support
Adaptability	Adapt → Learn → Grow
Learning	Small Goals → Practice → Projects
Accuracy	Check → Verify → Deliver
Final Answer	Commitment → Skills → Value

82. Salary Negotiation (UK £35–45k)

For your situation, salary negotiation is especially important because you're targeting London.

82.1 When should you discuss salary?

The safest order is:

1. HR screening.
2. Technical interview.
3. Final discussion.
4. Salary conversation.

Avoid becoming the first person to mention a number unless asked directly.

82.2 If HR asks first

What salary are you expecting?

82.2.1 Best Answer (B2)

Based on my research and the responsibilities of this role, I'm looking for something in the region of **£35,000 to £45,000**, depending on the overall package and growth opportunities.

Why this works:

- sounds flexible;
 - gives a range;
 - avoids sounding demanding.
-

82.3 If they offer £32,000

Don't reject immediately.

82.3.1 Good Response

Thank you for the offer.

I appreciate it.

Based on my research and my skills, I was expecting something closer to **£35,000**.

Is there any flexibility?

Professional.

Calm.

Confident.

82.4 If they ask for your minimum salary

Don't say:

"I'll take anything."

Instead:

I'm looking for a package that reflects the responsibilities of the role, ideally starting around **£35,000**.

82.5 Salary Negotiation Checklist

- Stay calm.
 - Pause before answering.
 - Give a range.
 - Mention market research.
 - Stay professional.
-

83. Recruiter Red Flags

Sometimes interviews reveal problems about the employer.

83.1 Green Flags

- Clear role expectations.
- Friendly communication.
- Learning opportunities.
- Structured interview.
- Honest answers.

83.2 Yellow Flags

- Unclear responsibilities.
- Frequent interruptions.
- Constantly changing answers.

Ask more questions.

83.3 Red Flags

- Nobody can explain the role.
- Unrealistic workload.
- Pressure to accept immediately.
- Negative comments about previous employees.
- No onboarding plan.

Remember:

You're interviewing the company too.

84. Mini Mock Interview

Practice without reading.

84.1 Interview Starts

Interviewer

Tell me about yourself.

(Pause.)

Answer.

Interviewer

Why Data Analytics?

(Pause.)

Answer.

Interviewer

Tell me about a challenge you solved.

(Pause.)

Answer.

Interviewer

Why should we hire you?

(Pause.)

Answer.

Interviewer

Do you have any questions for us?

(Pause.)

Answer.

85. Self-Evaluation

Score yourself.

Skill	Score
Confidence	1–5
Clarity	1–5
Structure	1–5
Natural English	1–5
Examples	1–5

Total:

___/25

Repeat until you consistently score above **20**.

86. Final Interview Checklist

86.1 The Day Before

- Research the company.
- Review your CV.
- Practise five key answers.
- Prepare questions.
- Charge your laptop.
- Check your internet.

86.2 One Hour Before

- Open the meeting link.
- Prepare water.
- Close distractions.
- Keep notes nearby.

86.3 During the Interview

- Smile.
- Speak slowly.
- Pause before answering.
- Use STAR.
- Ask questions.

86.4 After the Interview

Within 24 hours, send a short thank-you message.

Example:

Thank you for your time today.

I enjoyed learning more about the role and the team.

I appreciate the opportunity and look forward to hearing from you.

Professional.

Simple.

Effective.

87. Part 1 Complete

Congratulations.

You have now completed **HR Interview Mastery**.

Before moving to the SQL Bible, make sure you can confidently answer all **50 questions** without memorising every word.

The goal is sounding like a future colleague—not sounding like you’ve memorised a script. {callout-tip title=“Practice Before Your Interview”}

Before moving to the next chapter:

- Answer each question aloud in English.
- Keep each answer between **60–90 seconds**.
- Use the **STAR** framework whenever possible.
- Record yourself and listen for clarity and confidence.

The goal is not to memorise answers, but to communicate naturally.

II

179	LAG()	389
180	LEAD()	391
181	Running Total	393
182	Moving Average	395
183	Window Function Cheat Sheet	397
184	Business Case 1 — Tesco	399
185	Business Case 2 — Amazon	401
186	Business Case 3 — Deliveroo	403
187	Business Case 4 — NHS	405
188	Business Case 5 — Wise	407
189	Business Case 6 — Monzo	409
190	Business Case 7 — Retail Dashboard	411
191	Business Case 8 — Customer Retention	413
192	Business Case 9 — Product Performance	415
193	Business Case 10 — Executive Report	417
194	Full Mock SQL Interview (45 Minutes)	419
194.1	Part 1 — Warm-up (10 min)	419
194.2	Part 2 — Aggregations (10 min)	419
194.3	Part 3 — JOINS (10 min)	419
194.4	Part 4 — Advanced SQL (15 min)	419
195	Interview English Cheat Sheet	421
196	Common Interview Mistakes	423
197	5-Minute Emergency Revision	425
197.1	SELECT	425
197.2	WHERE	425
197.3	GROUP BY	425
197.4	HAVING	425
197.5	JOIN	425
197.6	CTE	425
197.7	ROW_NUMBER	425
197.8	RANK	425
197.9	DENSE_RANK	425
197.10	LAG	425
197.11	LEAD	425
198	Final SQL Challenge	427
199	SQL Bible Complete	429

Part II — SQL Bible

88. SQL Bible

89. Sprint 1 — Foundations

89.1 SELECT • WHERE • ORDER BY • LIMIT • DISTINCT

Welcome to the SQL Bible.

This book is designed for **Junior Data Analyst interviews in the UK**.

Instead of learning isolated syntax, you'll solve realistic business problems using one consistent dataset.

90. Introducing the Evenec Retail Dataset

Throughout the SQL Bible we'll work with one fictional company.

Evenec Retail

An online retail business selling products across the UK.

This makes interview practice much closer to real companies like Tesco, Amazon, Deliveroo or Wise.

90.1 Database Structure

Table	Purpose
customers	Customer information
orders	Orders placed
order_items	Products inside orders
products	Product catalogue
payments	Payment information
employees	Internal staff

91. Main Tables

91.1 customers

Column	Type
customer_id	INT
first_name	TEXT
last_name	TEXT
city	TEXT
signup_date	DATE

91.2 orders

Column	Type
order_id	INT
customer_id	INT
order_date	DATE
amount	DECIMAL
status	TEXT

91.3 products

Column	Type
product_id	INT
product_name	TEXT
category	TEXT
price	DECIMAL

92. Memory Map

Throughout this book remember:

Customer places an Order.

Order contains Items.

Items reference Products.

This simple relationship helps you understand joins later.

93. Question 1 — Show Every Customer

93.1 Recruiter Psychology

This tests whether you understand the most fundamental SQL operation.

93.2 Business Task

The manager wants to see every customer.

93.3 SQL Solution

```
SELECT *  
FROM customers;
```

93.4 Explain It (B2)

This query selects every column from the customers table.

93.5 Memory Hook

SELECT = Choose.

93.6 Common Mistake

Using `SELECT ALL`.

`SELECT *` already returns all columns.

93.7 Practice Challenge

Return only:

- first_name
 - city
-

94. Question 2 — Select Specific Columns

94.1 Recruiter Psychology

Interviewers prefer selecting only necessary columns.

94.2 Business Task

Display customer names and cities.

94.3 SQL Solution

```
SELECT first_name,  
       city  
FROM customers;
```

94.4 Explain It (B2)

Instead of returning every column, I only select the information I need.

94.5 Memory Hook

Only what matters.

94.6 Common Mistake

Using `SELECT *` in every query.

95. Question 3 — Filter London Customers

95.1 Recruiter Psychology

This checks your ability to filter data.

95.2 Business Task

Find customers living in London.

95.3 SQL Solution

```
SELECT first_name,  
       city  
FROM customers  
WHERE city = 'London';
```

95.4 Explain It (B2)

First I filter the rows where the city is London, then I return the required columns.

95.5 Memory Hook

WHERE filters rows.

95.6 Common Mistake

Using = incorrectly with text.

Always use quotes.

96. Quick Drill

Without looking:

Write a query that finds customers from Manchester.

97. Question 4 — Orders Above £100

97.1 Recruiter Psychology

Numeric filtering appears constantly during interviews.

97.2 Business Task

Show orders above £100.

97.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
WHERE amount > 100;
```

97.4 Explain It (B2)

I filter orders where the amount is greater than one hundred.

97.5 Memory Hook

Greater than means bigger money.

97.6 Common Mistake

Using `>=` when the question says “above”.

98. Question 5 — Multiple Conditions

98.1 Recruiter Psychology

Interviewers want to see logical operators.

98.2 Business Task

Find completed orders above £200.

98.3 SQL Solution

```
SELECT order_id,  
       amount,  
       status  
FROM orders  
WHERE amount > 200  
       AND status = 'Completed';
```

98.4 Explain It (B2)

Both conditions must be true.

98.5 Memory Hook

AND = Both.
OR = Either.

98.6 Common Mistake

Forgetting quotes around text.

99. Question 6 — OR Condition

99.1 Business Task

Find customers from London or Birmingham.

99.2 SQL Solution

```
SELECT first_name,  
       city  
FROM customers  
WHERE city = 'London'  
       OR city = 'Birmingham';
```

99.3 Explain It (B2)

Either city matches.

99.4 Memory Hook

OR opens another door. _____

100. Interview Tip

Many interviewers ask:

“Can you explain your query?”

Always explain:

1. Filter.
 2. Select.
 3. Result.
-

101. Question 7 — Sort Highest Revenue First

101.1 Recruiter Psychology

Sorting is one of the most common interview tasks.

101.2 Business Task

Show the largest orders first.

101.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
ORDER BY amount DESC;
```

101.4 Explain It (B2)

I sort the amounts from highest to lowest.

101.5 Memory Hook

DESC = Biggest first.

ASC = Smallest first.

101.6 Common Mistake

Forgetting DESC.

Default sorting is ascending.

102. Question 8 – Oldest Customers First

102.1 Business Task

Sort customers by signup date.

102.2 SQL Solution

```
SELECT first_name,  
       signup_date  
FROM customers  
ORDER BY signup_date ASC;
```

102.3 Explain It (B2)

The earliest customers appear first.

103. Question 9 — Top Five Orders

103.1 Recruiter Psychology

LIMIT appears frequently.

103.2 Business Task

Return only the five largest orders.

103.3 SQL Solution

```
SELECT order_id,  
       amount  
FROM orders  
ORDER BY amount DESC  
LIMIT 5;
```

103.4 Explain It (B2)

First I sort the orders, then I return only the top five.

103.5 Memory Hook

Sort.

Then limit.

104. Practice Challenge

Return the top three customers alphabetically.

105. Question 10 — Remove Duplicate Cities

105.1 Recruiter Psychology

DISTINCT is simple but frequently tested.

105.2 Business Task

List every city only once.

105.3 SQL Solution

```
SELECT DISTINCT city  
FROM customers;
```

105.4 Explain It (B2)

DISTINCT removes duplicate values.

105.5 Memory Hook

DISTINCT means unique.

105.6 Common Mistake

Using GROUP BY unnecessarily.

106. Question 11 — Find Products in Electronics

106.1 Business Task

Show products from the Electronics category.

106.2 SQL Solution

```
SELECT product_name,  
       price  
FROM products  
WHERE category = 'Electronics';
```

106.3 Explain It (B2)

I filter products by category.

107. Question 12 — Sort Products by Price

107.1 Business Task

Show the cheapest products first.

107.2 SQL Solution

```
SELECT product_name,  
       price  
FROM products  
ORDER BY price ASC;
```

107.3 Explain It (B2)

I sort products from the lowest price to the highest.

108. SQL Interview Vocabulary

Word	Meaning
Filter	Remove unwanted rows
Sort	Change order
Return	Show results
Column	Vertical field
Row	Single record

Use these words during interviews.

109. Mini Mock Interview

109.1 Interviewer

Show all customers.

Pause.

Answer.

109.2 Interviewer

Show completed orders above £200.

Pause.

Answer.

109.3 Interviewer

Return the five largest orders.

Pause.

Answer.

110. Cheat Sheet

110.1 SELECT

```
SELECT column  
FROM table;
```

110.2 WHERE

```
WHERE condition;
```

110.3 ORDER BY

```
ORDER BY column DESC;
```

110.4 LIMIT

```
LIMIT 5;
```

110.5 DISTINCT

```
SELECT DISTINCT column;
```

111. Memory Hooks Summary

Concept	Hook
SELECT	Choose
WHERE	Filter rows
ORDER BY	Organise
LIMIT	Stop early
DISTINCT	Unique

112. End-of-Sprint Challenge

Without looking at previous examples, solve these tasks.

1. Show customers from London.
2. Show orders above £150.
3. Show completed orders.
4. Show the five largest orders.
5. List unique cities.
6. Show Electronics products sorted by price.

Target:

- under **8 minutes**
- explain every query aloud in English.

Congratulations.

You have completed **SQL Sprint 1**. {.callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

113. SQL Bible

114. Sprint 2 — Aggregations

114.1 GROUP BY • HAVING • Aggregate Functions

Congratulations.

You already know how to retrieve data.

Now you'll learn how analysts **summarise business performance**, which is exactly what hiring managers expect from a Junior Data Analyst.

115. Why Aggregations Matter

Most business questions are not:

Show me every order.

Instead they sound like:

- Which city generates the most revenue?
- How many customers signed up last month?
- What is the average order value?
- Which category sells best?

All of these require aggregation.

116. Memory Map

Think of the process like making piles.

Rows become groups.

Each group gets a calculation.

Example:

City	Orders
London	25
Manchester	18
Leeds	9

Rows disappear.

Groups remain.

117. Aggregate Functions Overview

Function	Purpose
COUNT()	Count rows
SUM()	Total value
AVG()	Average
MIN()	Smallest
MAX()	Largest

Memory Hook:

Count • Sum • Average • Minimum • Maximum

118. Question 13 — Count All Orders

118.1 Recruiter Psychology

Interviewers often begin with COUNT() because it's simple but essential.

118.2 Business Task

How many orders exist?

118.3 SQL Solution

```
SELECT COUNT(*) AS total_orders  
FROM orders;
```

118.4 Explain It (B2)

COUNT returns the number of rows.

118.5 Memory Hook

COUNT counts records.

118.6 Common Mistake

Using COUNT(amount) instead of COUNT(*).

119. Question 14 — Count London Customers

119.1 Business Task

Count customers living in London.

119.2 SQL Solution

```
SELECT COUNT(*) AS london_customers
FROM customers
WHERE city = 'London';
```

119.3 Explain It (B2)

First I filter London customers, then I count them.

120. Question 15 — Total Revenue

120.1 Recruiter Psychology

Revenue questions appear constantly.

120.2 Business Task

Calculate total sales.

120.3 SQL Solution

```
SELECT SUM(amount) AS total_revenue  
FROM orders;
```

120.4 Explain It (B2)

SUM adds every order amount together.

120.5 Memory Hook

SUM builds money.

121. Question 16 — Average Order Value

121.1 Business Task

Find the average order value.

121.2 SQL Solution

```
SELECT AVG(amount) AS average_order  
FROM orders;
```

121.3 Explain It (B2)

AVG calculates the typical order value.

121.4 Common Mistake

Confusing average with total.

122. Question 17 — Largest Order

122.1 Business Task

Find the highest order.

122.2 SQL Solution

```
SELECT MAX(amount) AS largest_order  
FROM orders;
```

122.3 Explain It (B2)

MAX returns the highest value.

123. Question 18 — Smallest Order

123.1 SQL Solution

```
SELECT MIN(amount) AS smallest_order  
FROM orders;
```

123.2 Explain It (B2)

MIN finds the lowest value.

124. Understanding GROUP BY

Imagine these rows.

Customer	Amount
Alice	120
Alice	80
Bob	60

GROUP BY creates:

Customer	Revenue
Alice	200
Bob	60

Rows become summaries.

Memory Hook:

GROUP BY creates piles.

125. Question 19 — Revenue by Customer

125.1 Recruiter Psychology

This is one of the most common interview questions.

125.2 Business Task

Calculate revenue for each customer.

125.3 SQL Solution

```
SELECT customer_id,  
       SUM(amount) AS revenue  
FROM orders  
GROUP BY customer_id;
```

125.4 Explain It (B2)

I group orders by customer and calculate total revenue.

125.5 Common Mistake

Forgetting GROUP BY.

126. Question 20 — Orders by Status

126.1 Business Task

Count completed and cancelled orders.

126.2 SQL Solution

```
SELECT status,  
       COUNT(*) AS orders  
FROM orders  
GROUP BY status;
```

126.3 Explain It (B2)

Every status becomes one group.

127. Question 21 — Customers by City

127.1 Business Task

Count customers in each city.

127.2 SQL Solution

```
SELECT city,  
       COUNT(*) AS customers  
FROM customers  
GROUP BY city;
```

127.3 Explain It (B2)

Each city becomes one group.

128. GROUP BY Rule

Every selected column must be either:

- grouped
- aggregated

Example.

Correct.

```
SELECT city,  
       COUNT(*)  
FROM customers  
GROUP BY city;
```

Wrong.

```
SELECT city,  
       first_name  
FROM customers  
GROUP BY city;
```

Memory Hook:

Every passenger needs a seat.

129. Question 22 — Average Revenue by City

129.1 Business Task

Calculate average order value by city.

129.2 SQL Solution

```
SELECT city,  
       AVG(amount) AS average_revenue  
FROM customers  
JOIN orders USING(customer_id)  
GROUP BY city;
```

129.3 Explain It (B2)

I join customers and orders, group by city and calculate the average.

Don't worry about JOIN syntax yet.

Sprint 3 explains it fully.

130. HAVING

WHERE filters rows.

HAVING filters groups.

Memory Hook.

WHERE before GROUP.

HAVING after GROUP.

131. Question 23 — Cities with More Than Five Customers

131.1 Recruiter Psychology

WHERE vs HAVING is a classic interview trap.

131.2 SQL Solution

```
SELECT city,  
       COUNT(*) AS customers  
FROM customers  
GROUP BY city  
HAVING COUNT(*) > 5;
```

131.3 Explain It (B2)

I first create city groups, then remove groups with five or fewer customers.

132. Question 24 — Customers Spending Over £500

132.1 Business Task

Find customers whose total spending exceeds £500.

132.2 SQL Solution

```
SELECT customer_id,  
       SUM(amount) AS revenue  
FROM orders  
GROUP BY customer_id  
HAVING SUM(amount) > 500;
```

132.3 Explain It (B2)

HAVING filters the grouped results.

133. WHERE vs HAVING

WHERE	HAVING
Filters rows	Filters groups
Before grouping	After grouping

Interview Tip.

If you're calculating something like SUM or COUNT,
think HAVING.

134. Question 25 — Top Revenue Cities

134.1 Business Task

Show cities ranked by revenue.

134.2 SQL Solution

```
SELECT city,  
       SUM(amount) AS revenue  
FROM customers  
JOIN orders USING(customer_id)  
GROUP BY city  
ORDER BY revenue DESC;
```

134.3 Explain It (B2)

I calculate revenue for every city and sort from highest to lowest.

135. Business Case

Imagine you're interviewing at Tesco.
The hiring manager asks.

Which city performs best?

Your answer.

1. Join customers and orders.
2. Group by city.
3. Calculate revenue.
4. Sort descending.

Notice.

You're describing thinking.
Not just SQL.

136. Mini Mock Interview

136.1 Interviewer

How many customers do we have?

Pause.

Answer.

136.2 Interviewer

Which city has the highest revenue?

Pause.

Answer.

136.3 Interviewer

Which customers spent over £500?

Pause.

Answer.

137. Cheat Sheet

137.1 COUNT

```
COUNT(*)
```

137.2 SUM

```
SUM(amount)
```

137.3 AVG

```
AVG(amount)
```

137.4 GROUP BY

```
GROUP BY customer_id
```

137.5 HAVING

```
HAVING SUM(amount) > 500
```

138. Memory Hooks

Concept	Hook
COUNT	Count records
SUM	Build money
AVG	Typical value
GROUP BY	Make piles
HAVING	Filter groups

139. End-of-Sprint Challenge

Complete these without looking.

1. Count all customers.
2. Count London customers.
3. Calculate total revenue.
4. Find average order value.
5. Find the largest order.
6. Revenue by customer.
7. Customers by city.
8. Cities with more than five customers.
9. Customers spending over £500.
10. Top revenue cities.

Target.

- **12 minutes**
- explain every query aloud in English.

Congratulations.

You have completed **SQL Sprint 2**. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

140. SQL Bible

141. Sprint 3 — JOINS, CTEs & Subqueries

141.1 The Language of Relationships

Welcome to the most important chapter for Junior Data Analyst interviews.

Most real business questions require combining information from multiple tables.

Instead of memorising JOIN syntax, you'll learn to think like an analyst.

142. The Evenec Retail Relationship Map

Our dataset follows one business story.

- Customers place Orders.
- Orders contain Items.
- Items reference Products.
- Orders receive Payments.

142.1 Entity Relationship Overview

customers

orders

payments

order_items

products

customer_id order_id order_id product_id Memory Hook:

Customer → Order → Item → Product

143. Why JOIN Exists

Imagine two separate tables.

customers

customer_id	first_name
1	Alice
2	Bob

orders

order_id	customer_id	amount
101	1	120
102	2	60

A JOIN connects them.

Result.

first_name	amount
Alice	120
Bob	60

144. JOIN Mental Model

Instead of remembering syntax,
remember relationships.

Keys connect stories.

Primary Key.
Foreign Key.
Bridge.

145. Question 26 — INNER JOIN

145.1 Recruiter Psychology

This is one of the most common interview questions.

145.2 Business Task

Show every order together with the customer's name.

145.3 SQL Solution

```
SELECT c.first_name,  
       o.order_id,  
       o.amount  
FROM customers c  
INNER JOIN orders o  
ON c.customer_id = o.customer_id;
```

145.4 Explain It (B2)

I connect customers and orders using customer_id and return customer names with their orders.

145.5 Memory Hook

INNER = Both must exist. _____

146. INNER JOIN Visual

Customers Orders Only the overlap remains.

147. Question 27 — LEFT JOIN

147.1 Business Task

Show every customer, even if they never ordered.

147.2 SQL

```
SELECT c.first_name,  
       o.order_id  
FROM customers c  
LEFT JOIN orders o  
ON c.customer_id = o.customer_id;
```

147.3 Explain It (B2)

Every customer appears. Missing orders become NULL.

147.4 Memory Hook

Left keeps everyone.

This phrase will appear throughout the book.

148. LEFT JOIN Visual

Customers Orders Everything from the left table survives.

149. Interview Trap

Question.

Why is LEFT JOIN useful?

Good answer.

It allows me to keep every customer even when related records don't exist.

Avoid saying.

Because it joins left.

Explain the business value. _____

150. Question 28 — Find Customers Without Orders

150.1 SQL

```
SELECT c.first_name
FROM customers c
LEFT JOIN orders o
ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

150.2 Explain It (B2)

I keep every customer and filter those without matching orders.

Memory Hook.

NULL means “missing”.

151. Question 29 — Join Three Tables

Business questions often need multiple joins.

Example.

Customers.

Orders.

Payments.

```
SELECT c.first_name,  
       o.order_id,  
       p.payment_method  
FROM customers c  
JOIN orders o  
ON c.customer_id = o.customer_id  
JOIN payments p  
ON o.order_id = p.order_id;
```

Explain.

First I connect customers to orders, then orders to payments.

Think.

Step by step.

152. JOIN Order

Always ask yourself.

What is my starting table?

Then walk through relationships.

Customer

↓

Order

↓

Payment

Never jump randomly.

153. CTE — Common Table Expression

Interviewers love CTEs because they improve readability.
Memory Hook.

CTE = Temporary workspace.

Instead of writing one giant query,
build it in steps.

154. Question 30 — First CTE

Business Task.

Find high-value customers.

```
WITH customer_revenue AS (  
    SELECT customer_id,  
           SUM(amount) AS revenue  
    FROM orders  
    GROUP BY customer_id  
)  
SELECT *  
FROM customer_revenue  
WHERE revenue > 500;
```

154.1 Explain It (B2)

First I create a temporary table, then I filter it.

Memory Hook.

WITH creates a temporary workspace.

155. Why CTE Is Better

Instead of.

```
SELECT ...
```

inside another huge query,

split the problem.

Cleaner.

Easier to explain.

Interviewers appreciate readability.

156. Subqueries

A subquery is simply

a query inside another query.

Memory Hook.

Box inside a box.

157. Question 31 — Above Average Orders

Business Task.

Find orders larger than the average.

```
SELECT order_id,  
       amount  
FROM orders  
WHERE amount >  
(  
  SELECT AVG(amount)  
  FROM orders  
);
```

157.1 Explain It (B2)

First SQL calculates the average, then it compares every order against it.

158. Subquery Visual

Main Query

AVG(amount) One query feeds another.

159. Question 32 — Products Above Average Price

```
SELECT product_name,  
       price  
FROM products  
WHERE price >  
(  
    SELECT AVG(price)  
    FROM products  
);
```

Explain.

Products become filtered using the average price.

160. CTE vs Subquery

CTE	Subquery
Easier to read	Shorter
Reusable	Quick calculations
Better for interviews	Fine for simple tasks

Interview Tip.

If readability matters,
choose a CTE.

161. Business Case — Amazon Style

Imagine this question.

Find customers whose spending is above the company average.

Think aloud.

1. Calculate customer revenue.
2. Calculate average revenue.
3. Compare.

Interviewers score your reasoning,
not just the final SQL.

162. Mini Mock Interview

Interviewer.

Explain LEFT JOIN.

Pause.

Answer.

Interviewer.

Why use a CTE?

Pause.

Answer.

Interviewer.

Find orders above average.

Pause.

Answer.

163. Cheat Sheet

163.1 INNER JOIN

```
INNER JOIN table  
ON key = key;
```

163.2 LEFT JOIN

```
LEFT JOIN table  
ON key = key;
```

163.3 CTE

```
WITH temp AS (...)  
SELECT ...
```

163.4 Subquery

```
WHERE value >  
(  
  SELECT AVG(...)  
)
```

164. Memory Hooks Summary

Concept	Hook
INNER	Both exist
LEFT	Left keeps everyone
NULL	Missing
CTE	Temporary workspace
Subquery	Box inside a box

165. End-of-Sprint Challenge

Complete these without looking.

1. Join customers and orders.
2. Show every customer using LEFT JOIN.
3. Find customers without orders.
4. Join customers, orders and payments.
5. Build a CTE.
6. Find orders above average.
7. Find products above average price.
8. Explain INNER JOIN in English.
9. Explain LEFT JOIN in English.
10. Explain why CTE improves readability.

Target.

- **15 minutes**
- explain every solution aloud in English.

Congratulations.

You have completed **SQL Sprint 3**. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

166. SQL Bible

167. Sprint 4 — Window Functions & Interview Finale

167.1 ROW_NUMBER • RANK • DENSE_RANK • LAG • LEAD • OVER()

Welcome to the final SQL sprint.

By now you've learned:

- filtering;
- aggregation;
- joins;
- CTEs;
- subqueries.

Now you'll learn the functions that often separate a confident Junior Analyst from someone who only knows basic SQL.

168. Why Window Functions Matter

Imagine this question.

Who are the top three customers in every city?

GROUP BY cannot answer this alone.

Window functions calculate values **without destroying individual rows**.
Memory Hook.

GROUP BY collapses.

WINDOW looks through.

169. The Window Mental Model

Think of standing inside a moving window.

The table stays.

The calculation changes.

Rows remain visible.

170. OVER()

Every window function uses:

```
OVER(...)
```

Think of it as:

Look through this window.

171. Question 33 — ROW_NUMBER()

171.1 Recruiter Psychology

This is one of the most common technical interview questions.

171.2 Business Task

Number every order.

```
SELECT order_id,  
       amount,  
       ROW_NUMBER() OVER(ORDER BY amount DESC) AS row_num  
FROM orders;
```

171.3 Explain It (B2)

I sort orders by amount and assign a unique number to each row.

Memory Hook.

ROW_NUMBER never repeats.

172. Visual

Amount	ROW_NUMBER
500	1
400	2
300	3

Unique numbering.

173. Question 34 — RANK()

Business Task.

Rank customers by revenue.

```
SELECT customer_id,  
       revenue,  
       RANK() OVER(ORDER BY revenue DESC) AS customer_rank  
FROM customer_revenue;
```

Result.

Revenue	Rank
500	1
400	2
400	2
300	4

Notice.

Rank skips numbers.

Memory Hook.

RANK leaves gaps.

174. Question 35 — DENSE_RANK()

Same example.

```
SELECT customer_id,  
       revenue,  
       DENSE_RANK() OVER(ORDER BY revenue DESC) AS dense_rank  
FROM customer_revenue;
```

Result.

Revenue	Dense Rank
500	1
400	2
400	2
300	3

No gaps.

Memory Hook.

Dense fills gaps.

175. Interview Trap

Question.

When would you choose DENSE_RANK instead of RANK?

Good answer.

When tied values shouldn't create gaps in the ranking.

176. PARTITION BY

Window functions become much more powerful.

Example.

Rank customers **inside each city**.

```
SELECT customer_id,  
       city,  
       revenue,  
       ROW_NUMBER()  
       OVER(  
         PARTITION BY city  
         ORDER BY revenue DESC  
       ) AS city_rank  
FROM customer_revenue;
```

Memory Hook.

PARTITION creates mini competitions.

177. Visual

London.

Revenue	Rank
500	1
300	2

Manchester.

Revenue	Rank
600	1
250	2

Separate rankings.

178. Question 36 — Top Customer in Every City

Business Task.

Return only the best customer from each city.

```
WITH ranked AS
(
  SELECT customer_id,
         city,
         revenue,
         ROW_NUMBER()
           OVER(
             PARTITION BY city
             ORDER BY revenue DESC
           ) AS rn
  FROM customer_revenue
)

SELECT *
FROM ranked
WHERE rn=1;
```

Explain.

First I rank customers inside each city, then I keep only rank one.

179. LAG()

LAG compares with the previous row.

Business Task.

Compare today's revenue with yesterday's.

```
SELECT order_date,  
       revenue,  
       LAG(revenue)  
         OVER(ORDER BY order_date) AS previous_day  
FROM daily_sales;
```

Memory Hook.

LAG looks back.

180. LEAD()

LEAD looks forward.

```
SELECT order_date,  
       revenue,  
       LEAD(revenue)  
         OVER(ORDER BY order_date) AS next_day  
FROM daily_sales;
```

Memory Hook.

LEAD looks ahead.

181. Running Total

A classic interview question.

Business Task.

Calculate cumulative revenue.

```
SELECT order_date,  
       amount,  
       SUM(amount)  
       OVER(  
         ORDER BY order_date  
       ) AS running_total  
FROM orders;
```

Explain.

The total grows after every row.

Memory Hook.

Running total keeps growing.

182. Moving Average

Another popular interview task.

```
SELECT order_date,  
       revenue,  
       AVG(revenue)  
       OVER(  
         ORDER BY order_date  
         ROWS BETWEEN 6 PRECEDING  
         AND CURRENT ROW  
       ) AS seven_day_average  
FROM daily_sales;
```

Interview Tip.

Don't memorise.

Understand.

The window moves.

183. Window Function Cheat Sheet

Function	Purpose
ROW_NUMBER	Unique numbering
RANK	Ranking with gaps
DENSE_RANK	Ranking without gaps
LAG	Previous row
LEAD	Next row
SUM OVER	Running total

184. Business Case 1 — Tesco

Question.

Which city generated the most revenue?

Expected Thinking.

- Join customers.
 - Group by city.
 - Sum revenue.
 - Sort descending.
-

185. Business Case 2 — Amazon

Question.

Find the highest-spending customer in every city.

Expected Solution.

Use:

- CTE
 - ROW_NUMBER
 - PARTITION BY
-

186. Business Case 3 — Deliveroo

Question.

Find customers who haven't ordered recently.

Expected Thinking.

- Latest order.
 - Compare dates.
 - Filter inactive customers.
-

187. Business Case 4 — NHS

Question.

Calculate average appointments per patient.

Same SQL principle.

Business changes.

Logic stays.

188. Business Case 5 — Wise

Question.

Which payment method processes the most money?

Expected Thinking.

- Join payments.
 - Group by payment method.
 - SUM(amount).
-

189. Business Case 6 — Monzo

Question.

Find unusually large transactions.

Expected Thinking.

Use:

- AVG
 - Subquery
 - WHERE.
-

190. Business Case 7 — Retail Dashboard

Question.

Create KPIs.

Required.

- Total revenue
 - Average order
 - Largest order
 - Customer count
-

191. Business Case 8 — Customer Retention

Question.

Who made repeat purchases?

Expected Thinking.

Count orders.

HAVING COUNT()>1. _____

192. Business Case 9 — Product Performance

Question.

Top-selling category.

Expected Thinking.

Join.

Group.

Sort.

193. Business Case 10 — Executive Report

Imagine presenting to a CEO.
Don't say.

GROUP BY...

Instead say.

London generated the highest revenue, while Birmingham showed the fastest growth.

Analysts communicate business impact.

194. Full Mock SQL Interview (45 Minutes)

This simulates a real UK Junior Data Analyst interview.

194.1 Part 1 — Warm-up (10 min)

Interviewer

Show London customers.

Pause.

Answer.

Find completed orders.

Pause.

Answer.

Top five orders.

Pause.

Answer.

194.2 Part 2 — Aggregations (10 min)

Revenue by city.

Pause.

Answer.

Customers spending over £500.

Pause.

Answer.

Explain HAVING.

Pause.

Answer.

194.3 Part 3 — JOINS (10 min)

Connect customers and orders.

Pause.

Answer.

Find customers without orders.

Pause.

Answer.

Explain LEFT JOIN.

Pause.

Answer.

194.4 Part 4 — Advanced SQL (15 min)

Rank customers.

Pause.

Answer.

Above-average orders.

Pause.

Answer.

Top customer in every city.

Pause.

Answer.

Running total.

Pause.

Answer.

195. Interview English Cheat Sheet

Useful phrases.

Instead of:

“SQL does this...”

Say:

First I filter...

Then I group...

After that I calculate...

Finally I sort...

This sounds much more natural in interviews.

196. Common Interview Mistakes

✗ Writing before thinking.
✓ Explain your approach first.

✗ Memorising syntax only.
✓ Explain business reasoning.

✗ Panicking during joins.
✓ Draw relationships mentally.

197. 5-Minute Emergency Revision

Remember these.

197.1 **SELECT**

Choose columns.

197.2 **WHERE**

Filter rows.

197.3 **GROUP BY**

Create groups.

197.4 **HAVING**

Filter groups.

197.5 **JOIN**

Connect stories.

197.6 **CTE**

Temporary workspace.

197.7 **ROW_NUMBER**

Unique numbering.

197.8 **RANK**

Leaves gaps.

197.9 **DENSE_RANK**

No gaps.

197.10 **LAG**

Looks back.

197.11 **LEAD**

Looks ahead.

198. Final SQL Challenge

Complete these ten tasks without looking.

1. London customers.
2. Revenue by city.
3. Top five orders.
4. Customers over £500.
5. Join customers and orders.
6. Customers without orders.
7. Above-average orders.
8. Top customer per city.
9. Running total.
10. Rank customers.

Target.

- **25 minutes**
 - explain every query aloud in English.
-

199. SQL Bible Complete

Congratulations.

You have completed all four SQL sprints.

Before moving to Excel and Power BI, make sure you can:

- write queries without copying;
- explain every query in English;
- recognise business questions behind SQL tasks.

Remember.

Companies don't hire analysts because they know SQL.

They hire analysts because they solve business problems with SQL. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

III

273	Sprint 4 — Excel Assessment Centre	579
273.1	Dashboards • Business Cases • Mock Interview • 30-Minute Challenge	579
274	Assessment Overview	581
275	Before You Start	583
276	Assessment Task	585
277	Task 1 — Revenue KPI	587
278	Task 2 — Total Orders	589
279	Task 3 — Average Order Value	591
280	KPI Layout	593
281	Task 4 — Revenue by City	595
282	Task 5 — Best Selling Category	597
283	Task 6 — Interactive Dashboard	599
284	Dashboard Design Rules	601
285	Business Case 1 — CEO Report	603
286	Business Case 2 — Operations	605
287	Business Case 3 — Customer Growth	607
288	Business Case 4 — Executive Summary	609
289	Interview Question 11	611
290	Interview Question 12	613
291	Interview Question 13	615
292	Interview Question 14	617
293	Mini Mock Interview	619
294	The 30-Minute Assessment	621
295	Rebuild the Dashboard Yourself	623
296	Assessment Rubric	625
297	Recruiter Red Flags	627
298	B2 Presentation Script	629
299	Emergency Interview Cheat Sheet	631
300	Final Excel Challenge	633
301	Excel Bible Complete	635

Part III — Excel Interview Bible

200. Excel Interview Bible

201. Sprint 1 — Excel Foundations

201.1 Tables • Filters • Sorting • Essential Navigation

Welcome to the Excel Interview Bible.

Unlike traditional Excel courses, this chapter prepares you for **real Junior Data Analyst interviews in UK companies**.

We'll use the **Evenec Retail Playground** throughout every exercise.

202. The Evenec Retail Dataset

Open:

`playground/evenec-retail/csv/customers.csv`

This is your first working dataset.

You are already familiar with it from SQL.

Now we'll analyse it in Excel.

203. Why Companies Use Excel

SQL extracts data.

Excel explores it.

A typical workflow looks like this.

- SQL exports data.
- Excel cleans it.
- Pivot Tables summarise it.
- Charts communicate results.

Memory Hook.

SQL finds data.

Excel explains data.

204. Interview Question 1

204.1 Recruiter Psychology

Can you work confidently with raw spreadsheets?

204.2 Task

Open `customers.csv`.

Convert it into an Excel Table.

204.3 Steps

1. Select any cell.
2. Press **Ctrl + T**.
3. Tick "My table has headers."

204.4 Why Tables Matter

Tables automatically expand.

Formulas become easier.

Filters appear automatically.

204.5 Bogdan Notes

During interviews, creating a Table quickly signals confidence with Excel fundamentals.

205. Interview Question 2

205.1 Filters

Task.

Show only customers from London.

205.1.1 Steps

- Click the filter arrow.
- Choose London.

205.1.2 Interview English

I filtered the customer list to display only London records.

Memory Hook.

Filter hides.

Delete removes.

Never confuse them.

206. Interview Question 3

206.1 Multi-Level Sorting

Task.

Sort customers by:

1. City (A-Z)
2. Signup Date (Oldest First)

206.1.1 Steps

Data → Sort

Add another level.

206.1.2 Why Recruiters Ask This

Real reports rarely use one sort.

207. Freeze Panes

Imagine scrolling through 5,000 rows.

Headers disappear.

Solution.

View → Freeze Panes.

Memory Hook.

Freeze protects headers.

208. Interview Question 4

Find the newest customer.

Hint.

Sort by Signup Date.

Descending.

209. Essential Keyboard Shortcuts

Shortcut	Purpose
Ctrl+T	Create Table
Ctrl+Shift+L	Toggle Filters
Ctrl+Arrow	Jump through data
Ctrl+Home	Top of sheet
Ctrl+End	Last used cell

Practice until these become automatic.

210. Structured References

Instead of

```
=A2+B2
```

Tables allow

```
=[@Revenue]+[@Tax]
```

Interview Tip.

Recruiters often prefer structured references because they're easier to maintain.

211. Interview Question 5

Count customers using the status bar.

Don't write a formula.

Select the column.

Read the status bar.

This demonstrates practical Excel knowledge.

212. Removing Duplicates

Task.

Find unique cities.

Data → Remove Duplicates.

Compare this with SQL.

SQL used:

```
SELECT DISTINCT city  
FROM customers;
```

Notice.

Different tool.

Same business goal.

213. Find & Replace

Useful for cleaning imported data.

Shortcut.

Ctrl + H

Interview Example.

Replace

Ltd.

with

Limited.

214. Data Validation

Business Task.

Create a dropdown list for order status.

Possible values.

- Completed
- Pending
- Cancelled

Why it matters.

Prevents inconsistent data entry.

215. Conditional Formatting

Task.

Highlight customers who joined this year.

Steps.

Home → Conditional Formatting.

Interview English.

I used conditional formatting to highlight recent customers.

216. Quick Quality Checks

Before analysing any spreadsheet, ask:

- Missing values?
- Duplicate rows?
- Correct dates?
- Correct currency?
- Consistent spelling?

Memory Hook.

Clean before analysing.

217. Mini Mock Interview

Interviewer.

Convert this CSV into a Table.

Pause.

Answer.

Interviewer.

Show only London customers.

Pause.

Answer.

Interviewer.

Find duplicate cities.

Pause.

Answer.

218. Practice Challenge

Using `customers.csv`.

Complete these tasks.

1. Convert to Table.
2. Filter London.
3. Sort by city and date.
4. Freeze headers.
5. Find newest customer.
6. Remove duplicate cities.
7. Create a dropdown list.
8. Highlight recent customers.

Target.

10 minutes

Explain every action aloud in English.

219. Memory Hooks

Concept	Hook
Table	Smart spreadsheet
Filter	Hide rows
Sort	Change order
Freeze	Keep headers
Validation	Prevent mistakes

Congratulations.

You completed **Excel Sprint 1**. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

220. Excel Interview Bible

221. Sprint 2 — XLOOKUP, INDEX/MATCH & Text Functions

221.1 Find • Clean • Combine • Validate

Welcome to Sprint 2.

This chapter teaches the Excel functions that recruiters most often expect Junior Data Analysts to know.

We'll continue using the **Evenec Retail Playground**.

Main files:

- `customers.csv`
 - `orders.csv`
 - `products.csv`
-

222. Why Lookup Functions Matter

Imagine this business question.

“Find the customer’s city using only their Customer ID.”

This is exactly what lookup functions solve.

Memory Hook.

IDs connect stories.

Just like SQL JOIN.

223. Interview Question 6

223.1 XLOOKUP Basics

223.1.1 Recruiter Psychology

This is currently one of the most valuable Excel functions.

223.1.2 Business Task

Return the customer's city using `customer_id`.

223.1.3 Formula

```
=XLOOKUP(A2,Customers[customer_id],Customers[city])
```

223.1.4 Explain It (B2)

XLOOKUP searches for the customer ID and returns the matching city.

223.1.5 Memory Hook

Search.
Match.
Return.

223.1.6 Common Mistake

Mixing lookup and return columns.

224. Visual

Customer ID	Result
101	London
205	Leeds
412	Bristol

225. Why XLOOKUP Replaced VLOOKUP

VLOOKUP

XLOOKUP

Left-to-right only

Any direction

Column numbers

Column names

Less flexible

More flexible

Interview Tip.

If asked which you prefer,
choose XLOOKUP.

226. Interview Question 7

226.1 Product Price Lookup

Business Task.

Find product prices from `products.csv`.

Formula.

```
=XLOOKUP(B2,Products[product_id],Products[price])
```

Explain.

The product ID becomes the search key.

227. Missing Matches

Business Task.

Display “Not Found” instead of an error.

Formula.

```
=XLOOKUP(A2,Customers[customer_id],Customers[city],"Not Found")
```

Memory Hook.

Friendly errors improve reports.

228. INDEX + MATCH

Some companies still ask this.

Understand it.

Don't fear it.

Formula.

```
=INDEX(Customers[city],MATCH(A2,Customers[customer_id],0))
```

Explain.

MATCH finds the row.

INDEX returns the value.

Memory Hook.

MATCH finds.

INDEX retrieves.

229. XLOOKUP vs INDEX/MATCH

XLOOKUP	INDEX/MATCH
---------	-------------

Modern	Classic
--------	---------

Simpler	More technical
---------	----------------

Preferred	Still common
-----------	--------------

Interview Advice.

Know both.

Use XLOOKUP when available.

230. Interview Question 8

230.1 LEFT()

Business Task.

Extract the first three letters.

Formula.

```
=LEFT(B2,3)
```

Example.

London

↓

Lon

231. RIGHT()

Business Task.

Extract the last two characters.

Formula.

```
=RIGHT(B2,2)
```

Useful for codes.

232. MID()

Business Task.

Extract characters from the middle.

Formula.

```
=MID(B2,4,5)
```

Memory Hook.

Start.

Length.

Extract.

233. LEN()

Business Task.

Count characters.

Formula.

```
=LEN(B2)
```

Interview Example.

Find unusually short names. _____

234. TRIM()

One of the most practical functions.

Business Task.

Remove extra spaces.

Formula.

```
=TRIM(B2)
```

Why Recruiters Like This

Imported data often contains hidden spaces.

235. UPPER, LOWER, PROPER

Standardise text.

Upper.

```
=UPPER(B2)
```

Lower.

```
=LOWER(B2)
```

Proper.

```
=PROPER(B2)
```

Business Example.

Convert

john SMITH

into

John Smith.

236. TEXT()

Business Task.

Format dates.

Formula.

```
=TEXT(A2,"dd mmm yyyy")
```

Example.

2025-01-15

↓

15 Jan 2025

Interview Tip.

Formatting with TEXT doesn't change the original value.

237. CONCAT()

Business Task.

Build a full name.

Formula.

```
=CONCAT(B2," ",C2)
```

Alternative.

```
=B2&" "&C2
```

Memory Hook.

Join words.

238. TEXTJOIN()

Business Task.

Combine multiple values.

Formula.

```
=TEXTJOIN(" ", TRUE, A2:A5)
```

Very useful for reports.

239. Cleaning Imported Data

Real companies rarely receive perfect spreadsheets.

Check:

- Extra spaces
- Wrong capitalisation
- Duplicate IDs
- Blank cells

Always clean before analysing.

240. Interview Question 9

Customer Full Name

Using `customers.csv`.

Create.

John Smith

from:

John

•

Smith

Expected Formula.

```
=CONCAT([@first_name]," ",[@last_name])
```

241. Interview Question 10

Product Label

Create.

Electronics - £249.99

Formula.

```
=CONCAT([@category]," - £",[@price])
```

Business Purpose.

More readable reports.

242. Error Handling

Instead of showing

#N/A

use

```
=IFERROR(formula,"Missing")
```

Interview English.

IFERROR makes reports easier to understand.

Memory Hook.

Hide technical errors. _____

243. Mini Business Case

Imagine you're working at Deliveroo.
The manager asks.

Build a clean customer list.

Expected Workflow.

1. TRIM.
2. PROPER.
3. Remove duplicates.
4. Create Full Name.
5. Format dates.

Notice.

This is exactly how analysts think.

244. Mini Mock Interview

Interviewer.

Find a customer's city.

Pause.

Answer.

Interviewer.

Clean imported names.

Pause.

Answer.

Interviewer.

Explain XLOOKUP.

Pause.

Answer.

245. Formula Cheat Sheet

Task	Formula
Lookup	XLOOKUP
Alternative	INDEX/MATCH
First letters	LEFT
Last letters	RIGHT
Middle	MID
Length	LEN
Clean spaces	TRIM
Uppercase	UPPER
Lowercase	LOWER
Proper Case	PROPER
Format	TEXT
Join	CONCAT
Many values	TEXTJOIN

246. Memory Hooks

Function	Hook
XLOOKUP	Find & Return
MATCH	Find Position
INDEX	Bring Value
LEFT	Beginning
RIGHT	Ending
MID	Middle
TRIM	Clean Spaces
TEXT	Pretty Format

247. End-of-Sprint Challenge

Using the Evenec Retail CSV files, complete these tasks without copying the answers.

1. Lookup customer city.
2. Lookup product price.
3. Replace missing values with “Not Found”.
4. Build a full name.
5. Extract the first three letters.
6. Clean extra spaces.
7. Convert names to Proper Case.
8. Format dates.
9. Build a product label.
10. Explain XLOOKUP in English.

Target.

15 minutes

Remember.

Companies don’t hire people because they know formulas.

They hire people because they can quickly turn messy data into useful information.

Congratulations.

You completed **Excel Sprint 2**. {callout-tip title=“Practice with Real Data”}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

248. Excel Interview Bible

249. Sprint 3 — Manager’s Monday Morning Dashboard

249.1 Pivot Tables • Pivot Charts • Slicers • Timeline • Power Query

Monday.

09:00.

Your manager sends one message.

“I need a sales dashboard in twenty minutes.”

This chapter teaches exactly that workflow.

You’ll build a complete business dashboard using the **Evenec Retail Playground**.

Main files:

- orders.csv
 - customers.csv
 - products.csv
-

250. What We'll Build

By the end of this sprint you'll have a dashboard showing:

- Total Revenue
- Total Orders
- Average Order Value
- Top Performing City
- Best Selling Category
- Interactive Filters
- Timeline

This looks much closer to a real business report than a classroom exercise.

251. Dashboard Preview

Imagine this layout.

Revenue Orders Avg Order Top City

Revenue by City Category Sales

Timeline + Slicers Keep this mental picture while building.

252. Download the Real Dashboard

The dashboard in this chapter isn't just an illustration.
Open the real workbook from the Evenec Retail Playground.

[!TIP] Evenec Retail Executive Dashboard

Includes:

- KPI Cards
- Revenue Dashboard
- Pivot Charts
- Executive Layout



File:

`playground/evenec-retail/evenec_retail_dashboard.xlsx`

253. Step 1 — Import Data with Power Query

253.1 Recruiter Psychology

Many UK companies expect analysts to know basic Power Query.

253.2 Business Task

Import `orders.csv`.

253.2.1 Steps

1. Data
2. Get Data
3. From Text/CSV
4. Select `orders.csv`
5. Load to Excel

253.2.2 Explain It (B2)

I imported the CSV using Power Query instead of copying data manually.

Memory Hook.

Power Query is the kitchen.

It prepares ingredients before cooking.

254. Why Power Query Matters

Manual work:

- copy
- paste
- repeat

Power Query:

- refresh
- done

Interview Tip.

Mention refreshable workflows.
Recruiters like automation.

255. Step 2 — Build Your First Pivot Table

255.1 Business Task

Revenue by City.

255.1.1 Steps

Insert

→ Pivot Table

Fields.

Rows:

- city

Values:

- Sum of amount

255.1.2 Result

City	Revenue
London	£120,000
Manchester	£87,000

255.1.3 Explain It (B2)

The Pivot Table automatically groups cities and calculates total revenue.

Memory Hook.

Pivot means summarise.

256. Question 11

Why use a Pivot Table instead of formulas?

Good answer.

It's faster, easier to update and automatically groups data.

257. Step 3 — Sort Highest Revenue

Right-click.

Sort.

Largest to Smallest.

Now the manager immediately sees:

- top city
- weakest city

Business before beauty.

258. Step 4 — Create a Pivot Chart

Business Task.

Turn revenue into a chart.

Insert

→ Pivot Chart.

Choose.

Clustered Column.

Why this chart?

Easy comparison.

Interview English.

I chose a column chart because it's easier to compare categories.

259. Chart Psychology

Don't decorate.

Communicate.

Avoid:

- 3D charts
- unnecessary colours
- heavy shadows

Keep charts clean.

260. Step 5 — Revenue by Category

Use.

products.csv

and

orders.csv.

Create another Pivot Table.

Rows.

Category.

Values.

Revenue.

Business Question.

Which category performs best?

Exactly the type of question hiring managers ask.

261. Step 6 — Add Slicers

Business Task.

Filter instantly.

Insert

→ Slicer.

Choose.

- City
- Category

Now one click changes the whole dashboard.

Memory Hook.

Slicers are remote controls.

262. Step 7 — Add Timeline

Timeline works with dates.

Insert.

Timeline.

Choose.

Order Date.

Now the manager can switch between:

- month
- quarter
- year

This feels like a professional dashboard.

263. Step 8 — Build KPI Cards

Create four KPI boxes.

KPI	Formula
Revenue	SUM
Orders	COUNT
Average Order	AVERAGE
Top City	Pivot

Arrange them across the top.

Exactly like executive dashboards.

264. KPI Layout

Revenue Orders Average Top City Notice how little information creates a strong executive view.

265. Formatting Like a Professional

Use.

- white background
- dark text
- consistent spacing
- aligned cards

Good dashboards feel calm.

Not crowded.

266. Business Case — Monday Morning

The CEO asks.

Which city should receive more marketing budget?

Workflow.

1. Filter by month.
2. Compare cities.
3. Check category.
4. Present one recommendation.

Don't say.

"London has the highest SUM."

Say.

"London generated the highest revenue this month, so I'd recommend focusing additional marketing spend there."

Business language wins interviews.

267. Common Dashboard Mistakes

Avoid.

- too many colours
- tiny fonts
- inconsistent spacing
- duplicate charts
- manual calculations

Memory Hook.

One question.

One chart.

268. Mini Mock Interview

Interviewer.

Build revenue by city.

Pause.

Answer.

Interviewer.

Why choose a Pivot Table?

Pause.

Answer.

Interviewer.

Why add slicers?

Pause.

Answer.

269. Dashboard Checklist

Before presenting.

- ☐ Tables refreshed
- ☐ Filters working
- ☐ Slicers connected
- ☐ Timeline working
- ☐ KPI cards updated
- ☐ Charts readable
- ☐ No empty cells

Use this checklist before every interview assignment.

270. Power Query Cheat Sheet

Action	Purpose
Import CSV	Load data
Refresh	Update automatically
Remove Columns	Clean data
Change Type	Fix formats
Close & Load	Return to Excel
Memory Hook.	
Power Query prepares.	
Pivot Table explains.	

271. End-of-Sprint Challenge

Using the Evenec Retail Playground.

Build a complete dashboard.

Requirements.

- Revenue KPI
- Orders KPI
- Average Order KPI
- Revenue by City chart
- Revenue by Category chart
- City slicer
- Timeline
- Clean formatting

Target.

20 minutes

Explain every decision aloud in English.

Congratulations.

You completed **Excel Sprint 3**. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

272. Excel Interview Bible

273. Sprint 4 — Excel Assessment Centre

273.1 Dashboards • Business Cases • Mock Interview • 30-Minute Challenge

Welcome to the final Excel sprint.

Everything you've learned now comes together in one realistic hiring assessment.

Imagine you're interviewing for a **Junior Data Analyst** role at a UK company.

The interviewer sends one message.

"You have 30 minutes."

Your goal isn't just to use Excel.

Your goal is to make business decisions with data.

274. Assessment Overview

Time limit:

30 minutes

Dataset:

- customers.csv
- orders.csv
- products.csv

Deliverables.

- Clean workbook
- Dashboard
- Business insights
- Verbal explanation

Exactly what many UK companies expect.

275. Before You Start

Open.

`evenec_retail_dataset.xlsx`

or import the CSV files.

Check.

- Tables
- Dates
- Filters
- No missing formatting

Memory Hook.

Prepare before analysing.

276. Assessment Task

The Sales Director asks.

“Tell me how our business performed and where we should focus next.”

You have.

- 30 minutes.
 - One workbook.
 - One chance.
-

277. Task 1 — Revenue KPI

Create.

KPI	Goal
Total Revenue	Executive card

Business Question.

How much money did we generate?

Presentation Tip.

Large number.

Minimal text.

278. Task 2 — Total Orders

Build another KPI.

Use.

- Pivot Table

or

- COUNTA

Interview English.

This KPI shows the total number of completed orders.

279. Task 3 — Average Order Value

Formula.

```
=AVERAGE (amount )
```

Business Meaning.

How valuable is a typical customer purchase?

280. KPI Layout

Your dashboard should begin with four cards.

Revenue Orders Average Top City Keep the design simple.

Executives scan.

They don't read.

281. Task 4 — Revenue by City

Create.

Pivot Table.

Rows.

City.

Values.

Revenue.

Then create.

Pivot Chart.

Business Question.

Where should marketing invest more?

282. Task 5 — Best Selling Category

Use.

Products.

Orders.

Revenue.

Show.

Top category.

Interview Tip.

Don't just show numbers.

Explain why they matter.

283. Task 6 — Interactive Dashboard

Add.

- City slicer
- Timeline

Now demonstrate.

London.



Manchester.



Entire year.

This interaction impresses interviewers.

284. Dashboard Design Rules

Use.

- white background
- dark text
- aligned cards
- consistent spacing

Avoid.

- rainbow colours
- 3D charts
- unnecessary decorations

Professional dashboards feel calm.

285. Business Case 1 — CEO Report

Question.

Which city deserves more investment?

Expected Thinking.

1. Compare revenue.
2. Look for trends.
3. Recommend one action.

Good Answer.

London generated the strongest revenue, so I'd prioritise additional marketing there.

Notice.

Recommendation.

Not just observation. _____

286. Business Case 2 — Operations

Question.

Which category performs worst?

Expected Workflow.

- Pivot
- Sort
- Compare

Business Language.

This category may need promotional support.

287. Business Case 3 — Customer Growth

Question.

How has customer acquisition changed?

Use.

Signup Date.

Timeline.

Pivot.

Look for trends.

288. Business Case 4 — Executive Summary

Imagine presenting to the board.
Don't say.

"The Pivot Table shows..."

Instead.

Revenue increased, London remained our strongest market, and Electronics generated the highest sales.

Executives hear conclusions.
Not formulas.

289. Interview Question 11

Why choose a Pivot Table?

Model Answer (B2)

It's faster than building multiple formulas and it's easier to update when new data arrives.

290. Interview Question 12

When would you use Power Query?

Model Answer

I'd use Power Query when I need to clean or refresh imported data automatically.

291. Interview Question 13

Why use XLOOKUP?

Model Answer

It connects related data without manually copying information.

Notice.

Business language.

Always.

292. Interview Question 14

How do you check spreadsheet quality?

Model Answer

I check duplicates, missing values, date formats and incorrect text formatting.

Memory Hook.

Clean.

Then analyse.

293. Mini Mock Interview

Interviewer.

Show revenue by city.

Pause.

Answer.

Interviewer.

Why use slicers?

Pause.

Answer.

Interviewer.

Explain your dashboard.

Pause.

Answer.

294. The 30-Minute Assessment

Set a timer.

Follow this sequence.

Minutes	Task
0–5	Import & prepare
5–10	KPIs
10–15	Pivot Tables
15–20	Charts
20–25	Slicers & Timeline
25–30	Final review

Exactly how professionals work.

295. Rebuild the Dashboard Yourself

Now open:

`playground/evenec-retail/evenec_retail_dashboard.xlsx`

Don't just inspect it.

Recreate it from scratch.

295.0.1 Your Mission

- ☐ Rebuild KPI cards.
- ☐ Recreate both charts.
- ☐ Add Pivot Tables.
- ☐ Add Slicers.
- ☐ Add Timeline.
- ☐ Present your dashboard in English within 60 seconds.

This mirrors a real UK Excel assessment.

296. Assessment Rubric

Score yourself.

Skill	Points
Tables	10
Filtering	10
XLOOKUP	15
Pivot Tables	20
Charts	15
Dashboard	20
Presentation	10

Total.

100 points

Target.

80+

297. Recruiter Red Flags

Avoid.

- ✗ Overcomplicated dashboards.
- ✗ Tiny fonts.
- ✗ Manual calculations everywhere.
- ✗ No explanation.

Recruiters notice communication as much as Excel skills.

298. B2 Presentation Script

Imagine you've just finished.

Say.

I created an interactive dashboard showing revenue, order volume and customer performance. The Pivot Tables summarise the data automatically, while the slicers allow quick filtering by city and time period. Based on the results, I'd recommend focusing additional marketing on the strongest-performing city.

This sounds natural for B2 English.

299. Emergency Interview Cheat Sheet

Remember these.

Tool	Purpose
Table	Organise data
Filter	Focus
Sort	Prioritise
XLOOKUP	Connect
Pivot	Summarise
Power Query	Prepare
Slicer	Interact
Timeline	Time analysis

Memory Hook.

Prepare.

Summarise.

Explain.

300. Final Excel Challenge

Without copying any solutions.

Build a complete executive dashboard.

Requirements.

- Revenue KPI
- Orders KPI
- Average Order KPI
- Top City
- Revenue by City chart
- Category chart
- Slicer
- Timeline
- Clean formatting
- 60-second presentation

Target.

30 minutes

Explain every decision aloud in English.

301. Excel Bible Complete

Congratulations.

You completed all four Excel sprints.

You can now:

- clean imported data;
- build lookup formulas;
- create Pivot Tables;
- build executive dashboards;
- explain your work confidently in English.

Remember.

Companies don't hire analysts because they know Excel.

They hire analysts because they turn spreadsheets into business decisions. {callout-tip title="Practice with Real Data"}

Open the **Evenec Retail Playground** and run every query from this chapter on a real SQLite database.

Open on GitHub

[playground/evenec-retail](#)

IV

Part IV — Power BI Interview Bible

377	Assessment Overview	789
378	Before You Start	791
379	Assessment Scenario	793
380	Your Mission	795
381	Dashboard Blueprint	797
382	The 45-Minute Timer	799
383	Assessment Task 1	801
383.1	Revenue Overview	801
384	Assessment Task 2	803
384.1	City Performance	803
385	Assessment Task 3	805
385.1	Category Analysis	805
386	Assessment Task 4	807
386.1	Monthly Trend	807
387	Assessment Task 5	809
387.1	Interactive Experience	809
388	Assessment Task 6	811
388.1	Drill-through Page	811
389	Assessment Task 7	813
389.1	Custom Tooltips	813
390	Assessment Task 8	815
390.1	Navigation	815
391	Hiring Manager Questions	817
391.1	Question 1	817
391.2	Question 2	817
391.3	Question 3	817
391.4	Question 4	817
392	Recruiter Questions	819
392.1	Tell me about this dashboard.	819
392.2	What would you improve next?	819
393	Presentation Challenge	821
394	Assessment Rubric	823
395	Recruiter Red Flags	825
396	Final Checklist	827
397	Companion Files	829
398	Final Interview Mindset	831
399	Congratulations	833
400		835
401		837

302. Power BI Interview Bible

303. Sprint 1 — Power BI Foundations

303.1 Import • Data Model • Power Query • First Visuals

Welcome to the Power BI Interview Bible.

This isn't a button-by-button tutorial.

This chapter prepares you for **real Junior Data Analyst interviews in the UK**.

We'll continue working with the **Evenec Retail Playground**.

Main files:

- customers.csv
 - orders.csv
 - products.csv
 - payments.csv
 - employees.csv
-

304. Why Companies Use Power BI

Excel explains yesterday.

Power BI monitors today.

A typical workflow looks like this.

- SQL extracts data.
- Power Query cleans it.
- Power BI models it.
- Dashboards communicate decisions.

Memory Hook.

SQL finds.

Excel explores.

Power BI monitors.

305. Interview Question 1

305.1 Why Power BI?

Recruiter Psychology.

They're checking whether you understand business value.

Good Answer (B2).

Power BI allows companies to combine data from multiple sources and create interactive dashboards that update automatically.

Don't say.

It's like Excel.

Power BI is much bigger than Excel.

306. Step 1 — Create Your First Report

Open Power BI Desktop.

Choose.

Get Data

Select.

Text/CSV

Import.

- customers.csv
- orders.csv
- products.csv
- payments.csv
- employees.csv

Congratulations.

You've created your first report.

307. Power Query Opens Automatically

Notice.

Power Query appears before the data reaches Power BI.

Think of it as:

the kitchen before the restaurant.

Business data rarely arrives perfectly clean.

308. Interview Question 2

308.1 Power Query Basics

Task.

Open `customers.csv`.

Change.

- Date format
- Text format
- Remove unnecessary columns

Why?

Cleaning should happen before analysis.

Memory Hook.

Prepare.

Then analyse.

309. Close & Apply

After cleaning.

Click.

Close & Apply

Now the model updates.

Interview English.

I prefer cleaning data in Power Query because the process is repeatable.

Recruiters like repeatable workflows.

310. Step 2 — Build the Data Model

Open.

Model View.

You'll see multiple tables.

Now create relationships.

From	To
orders.customer_id	customers.customer_id
payments.order_id	orders.order_id

Think back to SQL.

These are JOIN relationships.

Memory Hook.

Power BI remembers relationships.

SQL recreates them every query.

311. Star Schema

Most companies use a Star Schema.

Orders

Customers

Products

Payments

Employees

Notice.

One fact table.

Several dimensions.

Exactly like professional BI systems.

312. Interview Question 3

Why is Star Schema useful?

Good Answer.

It makes reports easier to build and improves performance.

313. Step 3 — Create Your First Visual

Choose.

Bar Chart.

Fields.

- City
- Revenue

Immediately.

You have business insight.

314. Visual Types

Visual	Best For
Bar	Comparison
Line	Trends
Card	KPI
Table	Details
Map	Locations

Memory Hook.
One question.
One visual.

315. Step 4 — Create KPI Cards

Build.

- Total Revenue
- Total Orders
- Average Order

These become executive cards.

Exactly like Excel.

316. Dashboard Layout

Revenue Orders Average Top City Keep plenty of white space.
Professional dashboards breathe.

317. Step 5 — Add Filters

Power BI calls them.

Slicers

Add.

- City
- Category
- Date

Now the report becomes interactive.

Exactly what hiring managers expect.

318. Formatting Like a Professional

Use.

- white background
- dark text
- aligned visuals
- consistent spacing

Avoid.

- rainbow colours
- 3D effects
- unnecessary borders

Business before decoration.

319. Mini Business Case

The Sales Director asks.

Which city deserves more marketing budget?

Workflow.

1. Filter.
2. Compare.
3. Explain.

Good Answer.

London generated the highest revenue, so I'd recommend prioritising additional marketing there.

Notice.

Recommendation.

Not just numbers.

320. Mini Mock Interview

Interviewer.

Why use Power Query?

Pause.

Answer.

Interviewer.

Why create relationships?

Pause.

Answer.

Interviewer.

Why choose a Card visual?

Pause.

Answer.

321. Power BI Cheat Sheet

Tool	Purpose
Power Query	Clean
Model View	Relationships
Card	KPI
Bar Chart	Compare
Line Chart	Trend
Slicer	Filter
Memory Hook.	
Import.	
Model.	
Visualise.	
Explain.	

322. End-of-Sprint Challenge

Using the Evenec Retail Playground.

Complete these tasks.

- Import all CSV files.
- Clean data in Power Query.
- Create relationships.
- Build three KPI cards.
- Build one revenue chart.
- Add one slicer.

Target.

15 minutes

Explain every action aloud in English.

Congratulations.

You completed **Power BI Sprint 1**.

323. Power BI Interview Bible

324. Sprint 2 — Thinking in DAX

324.1 Measures • CALCULATE • FILTER • RELATED • ALL

Welcome to Sprint 2.

This chapter teaches the biggest mindset shift in Power BI.

DAX isn't Excel.

DAX thinks in **context**.

We'll continue using the **Evenec Retail Playground**.

Main files:

- customers.csv
 - orders.csv
 - products.csv
 - payments.csv
 - employees.csv
-

325. What Makes DAX Different?

Excel asks.

“What’s in this cell?”

DAX asks.

“Which rows are currently visible?”

Memory Hook.

Excel thinks in cells.

DAX thinks in context.

326. Understanding Context

Imagine.

Your report is filtered to:

- London
- January

When you calculate Revenue,

DAX automatically uses only those rows.

That's called **Filter Context**.

327. Two Types of Context

Context	Meaning
Row Context	Current row
Filter Context	Current visible data

Interview Tip.

Being able to explain these two concepts is more valuable than memorising dozens of functions.

328. Interview Question 4

Explain Filter Context.
Good Answer (B2).

Filter Context is the set of filters currently affecting a calculation, such as city, category or date.

Simple.
Professional.
Correct.

329. Your First Measure

Open.

Model View.

Create.

New Measure.

Write.

```
Total Revenue =  
SUM(Orders[amount])
```

Why a Measure?

Because it updates automatically whenever filters change.

Memory Hook.

Measures breathe.

Columns stay fixed.

330. Calculated Column vs Measure

Calculated Column	Measure
Stored	Calculated
Every row	Entire context
Static	Dynamic
Business Example. Customer Tier. → Column. Total Revenue. → Measure.	

331. Interview Question 5

Why use a Measure?

Model Answer.

Measures recalculate automatically whenever report filters change.

Recruiters love this answer.

332. COUNT vs COUNTROWS

Two similar functions.
Different purpose.

```
Total Orders =  
COUNTROWS(Orders)
```

Why COUNTROWS?
Because you're counting records.
Not values.
Memory Hook.
Rows.
Not cells.

333. AVERAGE

Create.

```
Average Order =  
AVERAGE(Orders[amount])
```

Now compare.

- Revenue
- Orders
- Average Order

You're building executive KPIs.

334. CALCULATE

This is the heart of DAX.

Formula.

```
London Revenue =  
CALCULATE(  
    [Total Revenue],  
    Customers[city]="London"  
)
```

Notice.

You're changing the context.

Memory Hook.

CALCULATE changes reality.

335. Why CALCULATE Matters

Without it.

Current filters remain.

With it.

You create new business questions.

Examples.

- London only
- Electronics only
- Last year only

This is why CALCULATE appears in almost every real project.

336. FILTER

Sometimes conditions become more complex.

Example.

```
Large Orders =  
CALCULATE(  
    [Total Revenue],  
    FILTER(  
        Orders,  
        Orders[amount]>200  
    )  
)
```

Business Meaning.

Revenue from high-value purchases.

Interview English.

FILTER allows me to create more specific business conditions.

337. RELATED

Power BI already knows relationships.

Now use them.

Example.

```
Customer City =  
RELATED(Customers[city])
```

Think back.

SQL.

```
JOIN
```

Excel.

```
XLOOKUP
```

Power BI.

```
RELATED
```

Same business goal.

Different tool.

338. RELATEDTABLE

Imagine.

You're inside Customers.

You want.

All related orders.

Use.

```
RELATEDTABLE(Orders)
```

Memory Hook.

RELATED looks outward.

RELATEDTABLE looks inward.

339. ALL

One of the most important interview functions.

Example.

```
Revenue % =  
DIVIDE(  
    [Total Revenue],  
    CALCULATE(  
        [Total Revenue],  
        ALL(Customers)  
    )  
)
```

Business Meaning.

Current revenue.

Compared to total company revenue.

Exactly what executives ask.

340. DIVIDE

Don't write.

```
A/B
```

Use.

```
DIVIDE(A,B)
```

Why?

- Safer.

- No divide-by-zero errors.

- Interview Tip.

- Using DIVIDE instead of / shows professional habits.

341. Time Intelligence Preview

Next sprint goes deeper.

For now.

Create.

```
Current Year Revenue =  
TOTALYTD(  
    [Total Revenue],  
    Orders[order_date]  
)
```

Notice.

Power BI understands dates differently from Excel.

342. Mini Business Case

The Sales Director asks.

How much revenue came from London?

Use.

CALCULATE.

Then answer.

London generated £X in revenue.

Business language matters.

343. Mini Business Case

Question.

What percentage of total revenue comes from Electronics?

Hint.

Use.

- CALCULATE
- ALL
- DIVIDE

Think before writing.

344. Common DAX Mistakes

Avoid.

- ✗ Using calculated columns instead of measures.
- ✗ Forgetting relationships.
- ✗ Ignoring Filter Context.
- ✗ Dividing manually.

Memory Hook.

Think.

Context.

Then formula.

345. DAX Cheat Sheet

Function	Purpose
SUM	Add values
COUNTROWS	Count records
AVERAGE	Average
CALCULATE	Change context
FILTER	Complex conditions
RELATED	Bring related value
RELATEDTABLE	Return related rows
ALL	Remove filters
DIVIDE	Safe division

Keep this table nearby.

346. Mini Mock Interview

Interviewer.

What's the difference between a Measure and a Column?

Pause.

Answer.

Interviewer.

Why use CALCULATE?

Pause.

Answer.

Interviewer.

What does ALL do?

Pause.

Answer.

347. End-of-Sprint Challenge

Using the Evenec Retail Playground.

Build these measures.

- Total Revenue
- Total Orders
- Average Order
- London Revenue
- Large Orders Revenue
- Revenue Percentage
- Current Year Revenue

Target.

20 minutes

Then explain each measure aloud in English.

348. Memory Hooks

Concept	Hook
Measure	Breathes
Column	Stored
CALCULATE	Changes reality
FILTER	More specific
RELATED	SQL JOIN
ALL	Remove filters
DIVIDE	Stay safe

Congratulations.

You completed **Power BI Sprint 2**.

349. Power BI Interview Bible

350. Sprint 3 — Executive Dashboard

350.1 KPI Cards • Slicers • Drill-through • Tooltips • Bookmarks

Monday.

09:00.

The Sales Director opens your dashboard.

Within ten seconds they should understand:

- How much revenue we generated.
- Which city performs best.
- Which category needs attention.
- What changed this month.

That's the purpose of executive dashboards.

We'll build this report using the **Evenec Retail Playground**.

Main files:

- customers.csv
 - orders.csv
 - products.csv
 - payments.csv
-

351. What We'll Build

By the end of this sprint you'll have:

- Executive KPI Cards
- Revenue Dashboard
- Category Analysis
- Interactive Slicers
- Drill-through Pages
- Custom Tooltips
- Navigation Buttons
- Professional Layout

This is much closer to a real company dashboard than a classroom exercise.

352. Dashboard Preview

Evenec Retail Executive Dashboard

Revenue Orders Average Top City

Revenue by City Revenue by Category

Interactive Slicers Keep this layout in mind while building.

353. Dashboard Design Principles

Good dashboards answer questions.

Bad dashboards display everything.

Follow these rules.

- White background.
- Dark text.
- Consistent spacing.
- Four KPI cards.
- Two main charts.
- Interactive filters.

Memory Hook.

One screen.

One story.

354. Step 1 — Build KPI Cards

Create these measures if they don't already exist.

```
Total Revenue =  
SUM(Orders[amount])
```

```
Total Orders =  
COUNTROWS(Orders)
```

```
Average Order =  
AVERAGE(Orders[amount])
```

Create four Card visuals.

- Revenue
- Orders
- Average Order
- Top City

Arrange them across the top.

Exactly like executive dashboards.

355. Top City Measure

Create.

```
Top City =  
VAR CityTable =  
    SUMMARIZE(  
        Customers,  
        Customers[city],  
        "Revenue",[Total Revenue]  
    )  
  
RETURN  
MAXX(  
    TOPN(1, CityTable, [Revenue]),  
    Customers[city]  
)
```

Business Meaning.

The dashboard always displays the strongest city.

356. Interview Question 6

Why use Cards?

Good Answer.

Cards allow executives to understand key metrics immediately.

Notice.

Business language.

Not technical language.

357. Step 2 — Revenue by City

Create.

Bar Chart.

Fields.

- City
- Total Revenue

Sort.

Highest first.

Now your manager instantly sees priorities.

358. Step 3 — Revenue by Category

Create another visual.

Rows.

Category.

Values.

Revenue.

Business Question.

Which product category deserves more investment?

Exactly the kind of question hiring managers ask.

359. Step 4 — Date Trend

Create.

Line Chart.

Axis.

Order Date.

Values.

Total Revenue.

Now trends become visible.

Memory Hook.

Bars compare.

Lines reveal change.

360. Step 5 — Slicers

Add.

- City
- Category
- Date

Now one click changes the entire dashboard.

Exactly what executives expect.

Interview English.

Slicers allow users to explore the data without changing the report itself.

361. Step 6 — Sync Slicers

Imagine multiple pages.

Users shouldn't repeat filters.

Open.

View.

Sync Slicers.

Now filters stay connected.

Professional reports use this constantly.

362. Step 7 — Drill-through

This is one of the strongest interview features.

Business Task.

Right-click London.

Open.

City Details.

Create a new page.

Name.

City Details

Add.

- Revenue
- Orders
- Category breakdown

Now users can investigate deeper.

Memory Hook.

Overview.

Then detail.

363. Step 8 — Custom Tooltips

Instead of tiny tooltips.

Create.

Tooltip Page.

Add.

- Revenue
- Orders
- Average Order

Now hovering over London reveals extra insights.

This feels premium.

364. Step 9 — Bookmarks

Bookmarks create interactive experiences.

Example.

Buttons.

- Overview
- Sales
- Categories

Each button switches views.

Business dashboards often use this technique.

365. Navigation Buttons

Insert.

Buttons.

Connect them to Bookmarks.

Now the report behaves like an application.

Interview Tip.

Navigation makes reports feel polished.

366. Conditional Formatting

Highlight.

Top performers.

Example.

Green.

High revenue.

Business Purpose.

Direct attention.

Don't make users search.

367. Business Case

The CEO asks.

Why did revenue fall?

Workflow.

1. Filter month.
2. Compare categories.
3. Drill through.
4. Explain findings.

Good Answer.

Revenue declined because one category underperformed while London remained stable.

Notice.

Insight.

Not description.

368. Storytelling with Data

Don't present.

Five charts.

Present.

One story.

Example.

Revenue grew.

Electronics led.

London remained strongest.

Three sentences.

Executive ready.

369. Mini Mock Interview

Interviewer.

Why use Drill-through?

Pause.

Answer.

Interviewer.

Why create Tooltips?

Pause.

Answer.

Interviewer.

Why sync slicers?

Pause.

Answer.

370. Executive Dashboard Checklist

Before presenting.

- ☐ KPI cards updated.
- ☐ Charts sorted.
- ☐ Slicers working.
- ☐ Drill-through tested.
- ☐ Tooltips working.
- ☐ Navigation buttons working.
- ☐ White space preserved.

Use this checklist before every interview assignment.

371. Power BI Dashboard Cheat Sheet

Feature	Purpose
Card	Executive KPI
Bar Chart	Compare
Line Chart	Trend
Slicer	Filter
Sync Slicers	Cross-page filtering
Tooltip	Extra insight
Bookmark	Interactive navigation
Drill-through	Detailed analysis
Memory Hook.	
Filter.	
Explore.	
Explain.	

372. End-of-Sprint Challenge

Build a complete Executive Dashboard.

Requirements.

- Revenue KPI
- Orders KPI
- Average Order KPI
- Top City KPI
- Revenue by City chart
- Revenue by Category chart
- Monthly Revenue trend
- City slicer
- Category slicer
- Date slicer
- Drill-through page
- Tooltip page
- Navigation buttons

Target.

25 minutes

Then present your dashboard in English within **60 seconds**.

373. Presentation Script (B2)

Imagine you've just finished.
Say.

I created an executive dashboard that shows revenue, order volume and customer performance. The KPI cards provide an immediate overview, while the slicers allow users to explore different cities and categories. I also added drill-through pages and custom tooltips to help managers investigate specific business questions without creating additional reports.

This sounds natural for a B2-level interview.

374. Memory Hooks

Concept	Hook
Card	Immediate insight
Slicer	Remote control
Drill-through	Go deeper
Tooltip	Hidden insight
Bookmark	Interactive app
White Space	Professional look

Congratulations.

You completed **Power BI Sprint 3**.

375. Power BI Interview Bible

376. Sprint 4 — Power BI Assessment Centre

376.1 45-Minute Challenge • Executive Presentation • Mock Interview

Congratulations.

You've reached the final sprint.

Everything you've learned now comes together in one realistic hiring assessment.

Imagine you're interviewing for a **Junior Data Analyst** role in London.

The hiring manager sends one message.

"You have forty-five minutes."

This chapter simulates a real UK assessment.

No memorisation.

No perfect conditions.

Just business decisions.

377. Assessment Overview

Time limit:

45 minutes

Dataset:

- customers.csv
- orders.csv
- products.csv
- payments.csv
- employees.csv

Deliverables.

- Clean Power BI report
- Interactive dashboard
- Business insights
- Executive presentation

Exactly what many UK companies expect.

378. Before You Start

Open.

Evenec Retail Playground

Import the CSV files.

Check.

- Relationships
- Date formats
- Measures
- Slicers

Memory Hook.

Prepare.

Then analyse.

379. Assessment Scenario

The Sales Director says.

“Our quarterly board meeting starts soon.”

Your dashboard must answer.

- How much revenue did we generate?
- Which city performs best?
- Which category needs attention?
- How should we invest next quarter?

One report.

One opportunity.

380. Your Mission

Build a complete executive dashboard.

Required.

- Revenue KPI
- Orders KPI
- Average Order KPI
- Top City KPI
- Revenue by City
- Revenue by Category
- Monthly Trend
- Slicers
- Drill-through
- Tooltips
- Navigation

Everything should feel polished.

381. Dashboard Blueprint

Evenec Retail Executive Dashboard

Revenue Orders Average Top City

Revenue by City Revenue by Category

Interactive Filters & Navigation Keep this structure while building.

382. The 45-Minute Timer

Follow this sequence.

Time	Task
0–5	Import & Model
5–12	Power Query Cleaning
12–20	Measures
20–28	Visuals
28–35	Slicers & Drill-through
35–40	Formatting
40–45	Presentation Review

Notice.

Professionals work against the clock.

383. Assessment Task 1

383.1 Revenue Overview

Build.

- Total Revenue
- Total Orders
- Average Order

Interview English.

I started by creating executive KPIs because they provide an immediate overview.

Business first.

384. Assessment Task 2

384.1 City Performance

Question.

Which city deserves additional marketing budget?

Expected Workflow.

- Bar Chart
- Sort descending
- Explain recommendation

Don't stop at numbers.

Give advice.

385. Assessment Task 3

385.1 Category Analysis

Question.

Which category needs attention?

Use.

- Bar Chart
- Slicer
- Drill-through

Business Answer.

This category contributes less revenue and may require promotional support.

386. Assessment Task 4

386.1 Monthly Trend

Question.

Is revenue becoming stronger or weaker?

Use.

- Line Chart
- Date hierarchy
- Month analysis

Executives love trends.

387. Assessment Task 5

387.1 Interactive Experience

Required.

- City slicer
- Category slicer
- Date slicer
- Sync Slicers

Then test everything.

Nothing should break.

388. Assessment Task 6

388.1 Drill-through Page

Create.

City Details.

Include.

- Revenue
- Orders
- Category breakdown

Now executives can investigate deeper.

Memory Hook.

Overview.

Then detail.

389. Assessment Task 7

389.1 Custom Tooltips

Create a Tooltip Page.

Show.

- Revenue
- Orders
- Average Order

Now hovering becomes informative.

Professional reports use this constantly.

390. Assessment Task 8

390.1 Navigation

Create.

Buttons.

- Overview
- Sales
- Categories

Use Bookmarks.

Now the report feels like an application.

391. Hiring Manager Questions

These often appear after the practical task.

391.1 Question 1

Why did you use Power Query?

Model Answer.

It creates a repeatable cleaning process instead of manual editing.

391.2 Question 2

Why build relationships instead of merging everything?

Model Answer.

A proper data model improves performance and makes reports easier to maintain.

391.3 Question 3

Why create Measures instead of Calculated Columns?

Model Answer.

Measures respond dynamically to filters, while calculated columns remain fixed after loading.

391.4 Question 4

Why choose a Card visual?

Model Answer.

Executives need immediate visibility of the most important KPIs.

392. Recruiter Questions

These test communication. _____

392.1 Tell me about this dashboard.

Good Answer (B2).

This dashboard helps managers monitor sales performance by city and category. It combines interactive filters with KPIs so decisions can be made quickly.

392.2 What would you improve next?

Example.

I'd add year-over-year comparisons and customer retention metrics.

Recruiters like improvement thinking.

393. Presentation Challenge

You now have.

60 seconds.

Imagine presenting to the board.

Say.

Good morning. This dashboard gives a quick overview of our sales performance. Revenue remains strongest in London, while Electronics continues to be our best-performing category. The slicers allow managers to explore different time periods and cities without changing the report. Based on these results, I'd recommend increasing marketing investment in our strongest regions while investigating weaker categories.

This sounds natural for B2 English.

394. Assessment Rubric

Score yourself.

Skill	Points
Import	5
Power Query	10
Relationships	10
Measures	20
Dashboard Design	20
Interactivity	15
Business Insights	10
Presentation	10

Total.

100 points

Target.

80+

Excellent.

90+

Interview ready.

395. Recruiter Red Flags

Avoid.

- ✗ Too many colours.
- ✗ Tiny text.
- ✗ Broken filters.
- ✗ No business explanation.
- ✗ Reading formulas instead of explaining decisions.

Remember.

Companies hire communicators.

Not formula collectors.

396. Final Checklist

Before submitting.

- ☐ Relationships correct.
- ☐ Measures working.
- ☐ Slicers tested.
- ☐ Drill-through working.
- ☐ Tooltips working.
- ☐ Navigation working.
- ☐ White space preserved.
- ☐ Presentation rehearsed.

Treat this checklist like a pilot before takeoff.

397. Companion Files

Practice using the Evenec Retail Playground.

Practice with Real Data

Open the companion files and recreate this assessment from scratch.

Included resources.

- `evenec_retail.db`
- `customers.csv`
- `orders.csv`
- `products.csv`
- `evenec_retail_dashboard.xlsx`

Complete the full challenge before checking your previous work.

398. Final Interview Mindset

Throughout this book you've learned.

Tool	Purpose
SQL	Extract
Excel	Analyse
Power BI	Communicate

This is how real analysts work.

Not one tool.

One business problem.

Solved across multiple tools.

399. Congratulations

You've completed.

- HR Interview Mastery
- SQL Bible
- Excel Bible
- Power BI Bible

More importantly.

You've practised thinking like a Data Analyst.
Welcome to the Evenec Interview Bible.

400.

401.